Experiments and Results — complete dump

Compiled 2026-07-27. Covers everything from the start of the internship (2026-05-15) to now.

Purpose. This is a *dump*, not an argument. It exists so collaborators can see what has actually been run, what numbers came out, where the artifacts live, and what is missing — and then decide what else needs to be there. Interpretation is kept to the minimum needed to make a number legible; the argument lives in the Master Collaborative Reference and the TMLR draft.

Sources. Three, cross-checked against each other:

1. `docs/Master_Collaborative_Reference_2026-07-27/` (chronicle + ledgers)
2. **The `/data` stores on this host** (`crfm-helm-audit`, `crfm-helm-audit-store`, `crfm-helm-public`, `dkps-embeddings`) — every number below marked *(disk)* was read off these directly today, not copied from prose
3. `docs/internship-powerpoints.zip` — seven weekly decks, 2026-06-02 ... 2026-07-14

Where the three disagree, §8 says so explicitly. **Four of those disagreements matter and are listed there — read §8 before citing anything.**

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1. Inventory at a glance

1.1 Execution totals *(disk)*

item	count
Experiment directories under `/data/crfm-helm-audit`	74
Local HELM runs with complete raw artifacts (`run_spec.json` + `scenario_state.json`)	**1,634**
Deployment-match sweep directories	26 (19 with a scored ranking)
Open-judge rejudge attempts (all `DONE`)	75 (63 full + 12 smoke)
Virtual experiments with a built aggregate report	12
Per-run core-metric reports	703 (Qwen) + 149 (OLMo) + 4 (GPT-OSS) + 29 early-experiment sets
Plots on disk across all reports	~8,900 PNG/JPG

1.2 Experiments by thread

#	Experiment	Kind	Date	Runs	Report?	Headline
A. HELM reproduction						
A1	Corpus cartography	Analysis	06-09 -> 06-17	34,512 records	[yes]	1,109 of 34,512 runs eligible
A2	phi-2 e2e instrument validation	Reproduction	05-27 -> 06-29	5 variants	[yes]	0.997 vLLM / 0.990 HF / 0.778 negative control
A3	OLMo campaign (combined)	Reproduction	06-12 -> 07-14	225 jobs	[yes]	0.935 mean agreement; ifeval +0.10 outlier

A4	Qwen reproduction (combined)	Reproduction	-> 07-20	775 jobs	[yes]	**0.975 mean agreement, 159 runs exact**
A5	GPT-OSS-20B from-spec	Reproduction	07-11 -> 07-14	12 jobs	[yes]	0.850 mean; drift ~25x tighter than OLMo ifeval
A6	Era-pinned classic (RedPajama)	Reproduction	07-08 -> 07-13	2 eras x 2 runs	[yes]	mmlu 1.000, synthetic_reasoning 0.817, both eras identical
A7	Deployment-match sweeps	Probe	07-07 -> 07-23	19 scored sweeps	tables	fp32 wins every unpinned-dtype family tested
A8	fp32 substrate closure	Probe + e2e	07-21 -> 07-23	2 e2e + 4 probes	tables	**539/541 byte-identical at full N**
A9	Qwen3.5-9B-Base extension	Compute	07-16 -> 07-17	72 runs	[no] **no report**	MMLU 0.810 (57 subj); 52% `<think>` leakage on boolq
A10	Early / pre-split experiments	Mixed	04-28 -> 05-20	~350 runs	partial	mostly superseded; 3 grids have reports
B. Open judges						
B1	Identity-replay gate	Reconstruction	07-18	6 benchmarks	[yes]	**6/6 exact, max err 1.95e-14**
B2	Judge substitution v1	Substitution	07-19	2 judges x 2 bench x 3 reps	[yes]	kappa 0.936 vs official's own 0.829
B3	Judge-size sweep	Substitution	07-20 -> 07-21	63 attempts	[!] stale	parse rate tracks size, calibration does not
C. Efficient benchmarking						
C1	mRMR tutorial replication	Method	06-02 -> 06-16	4 datasets	decks only	reproduced the paper's method ranking
C2	DKPS vs baselines	Method	06-16 -> 06-23	med_qa	decks only	DKPS loses at 20 models, closes with more

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2. Thread A — HELM reproduction

A1. Corpus cartography (Stage 1 filtering)

What. Crawl the public HELM mirror, extract per-run metadata, apply eligibility filters, emit a typed-reason census. **Artifact.** `/data/crfm-helm-audit-store/analysis/filter_inventory.json` (80 MB, generated 2026-06-17). Also reported in the 06-09 deck.

Results *(disk, recomputed today)*.

quantity	value

Total run records	**34,512**
Selected (eligible)	**1,109**
Excluded	33,403
Distinct models in the universe	366
Distinct benchmarks in the universe	185
Distinct models in the selected set	**33**
Distinct benchmarks in the selected set	**61**
Structurally incomplete	14
Eligible but out of scope	30

Exclusion reasons (runs can carry several):

reason	runs
no-local-helm-deployment	29,997
not-open-access	18,781
excluded-tags	17,526
too-large (>10B params)	7,624
not-text-like	6,024
missing-model-metadata	2,894
requires-closed-judge	484
requires-gated-dataset	74
structurally-incomplete	14

By public track (universe -> selected): classic 8,377->159 · lite 2,548->167 · mmlu 4,565->399 · mmlu-winogrande-afr 1,012->220 · thaiexam 140->70 · medhelm 871->56 · capabilities 385->12 · safety 431->4 · vhelm 9,873->0 · heim 1,992->0 · audio 1,737->0.

Grid coverage (06-09 deck): full grid 78,690 points (366 models x 215 scenarios), 6,844 populated; filtered grid 2,013 points (33 x 61), 198 populated.

> [!] **Every eligibility number in the paper traces to this one file, which is dated 2026-06-17 and is not regenerated from a checked-in manifest.** It is the single most-cited and least-refreshed artifact in the project. See §8.2 for a figure it contradicts.

A2. phi-2 end-to-end instrument validation

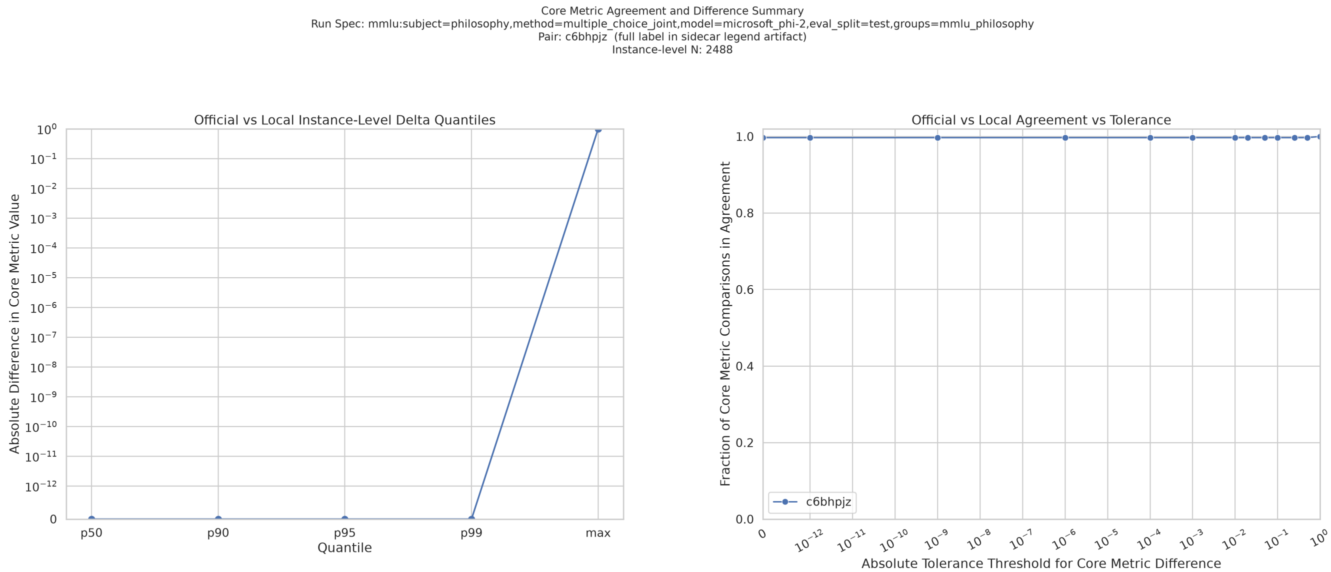
What. Deliberately-controlled reproduction of one public run (`mmlu:subject=philosophy,model=microsoft/phi-2`) under four deployments, plus one **negative control** (temperature 1) that *should* fail. This is the audit instrument validating itself. **Artifacts.** `/data/crfm-helm-audit-store/virtual-experiments/e2e-phi2*` (5 report trees), runs under `/data/crfm-helm-audit/e2e-phi_2-*`.

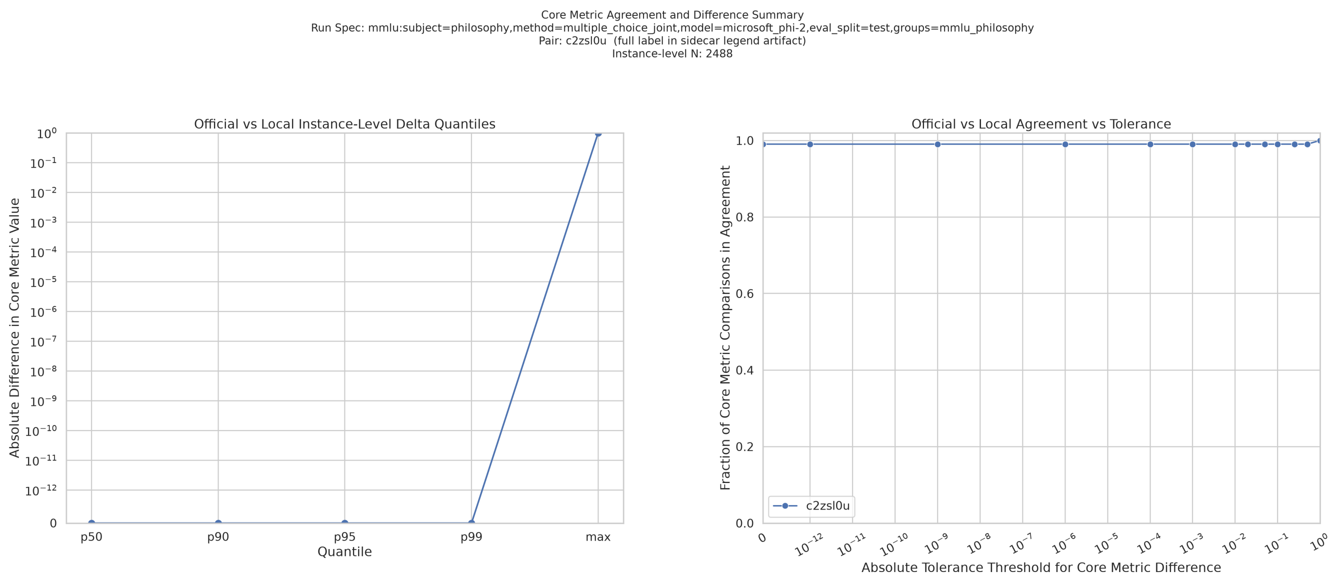
Results *(disk)* — agreement ratio at abs_tol=0:

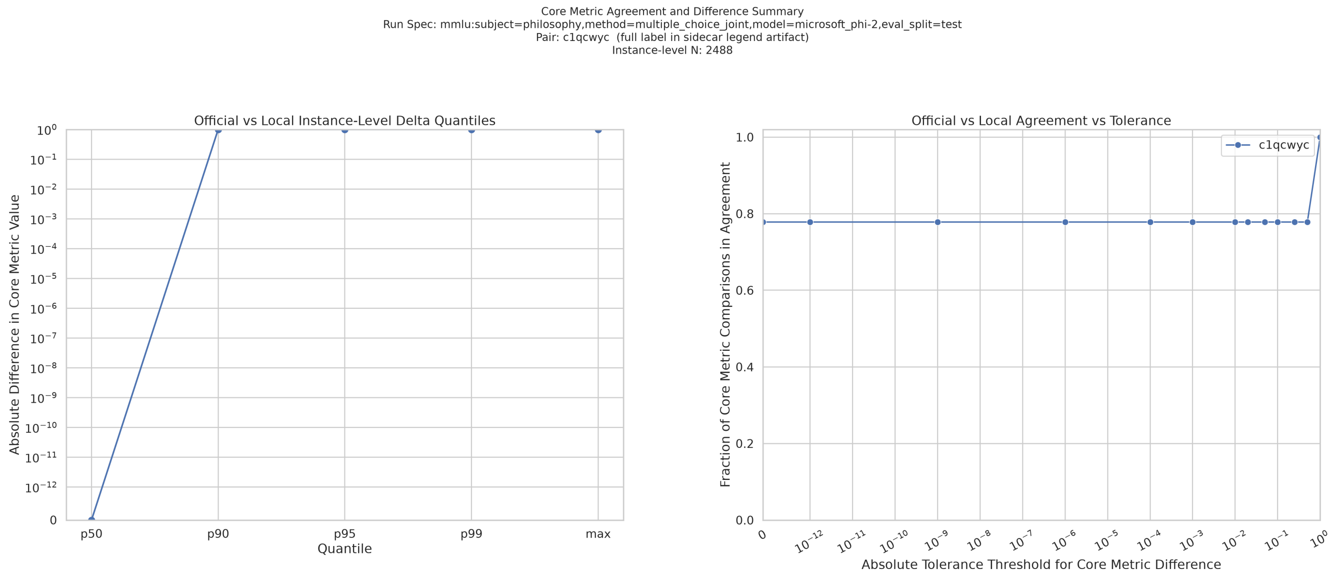
variant	deployment	agreement	bucket
`e2e-phi2-vllm`	vLLM, temp 0	**0.997**	high

`e2e-phi2-container`	vLLM in container, temp 0	**0.994**	high
`e2e-phi2-hf`	HuggingFace, temp 0	**0.990**	high
`e2e-phi2` (first pass)	8 jobs, 1 analyzed	0.804	moderate
`e2e-phi2-incomparable`	vLLM, **temp 1** (negative control)	**0.778**	low [yes] *expected*

The control working is the point: the instrument separates a reproduction from a deliberately-broken one, and the container adds ~0.003 rather than changing the verdict.







Three instrument bugs this found and fixed (06-02 deck): hard-coded infer-stack ports invisible to HELM; the `groups=` flag blocking otherwise-comparable public runs from matching; HELM stripping `temperature` from the run *name* so temp-bearing runs could not be located.

A3. OLMo campaign

What. Six OLMo models against their public HELM runs. The project's central case study and the vehicle for most of the machinery. **Artifacts.** `virtual-experiments/olmo-models-combined` (report regenerated 2026-07-14 16:25 UTC); runs under `/data/crfm-helm-audit/audit-allenai-olmo-*` (**73 + 114 + 114 + 76 + 33 + 6x4 runs with full raw artifacts on disk**).

Coverage *(disk)*. 225 jobs -> 220 with run artifacts -> 149 analyzed. 5 failures, all `unknown_failure`. All on `aiq-gpu`.

Reproducibility buckets: high (>=0.95) 79 · moderate (>=0.80) 56 · exact 7 · low 7.

Agreement at abs_tol=0 *(disk, 149 runs)*: mean **0.935**, median 0.952, min 0.673, 7 runs at 1.000. Relaxing to abs_tol=0.1 moves the mean only to 0.939 — **the disagreement is structural, not numerical-tolerance.**

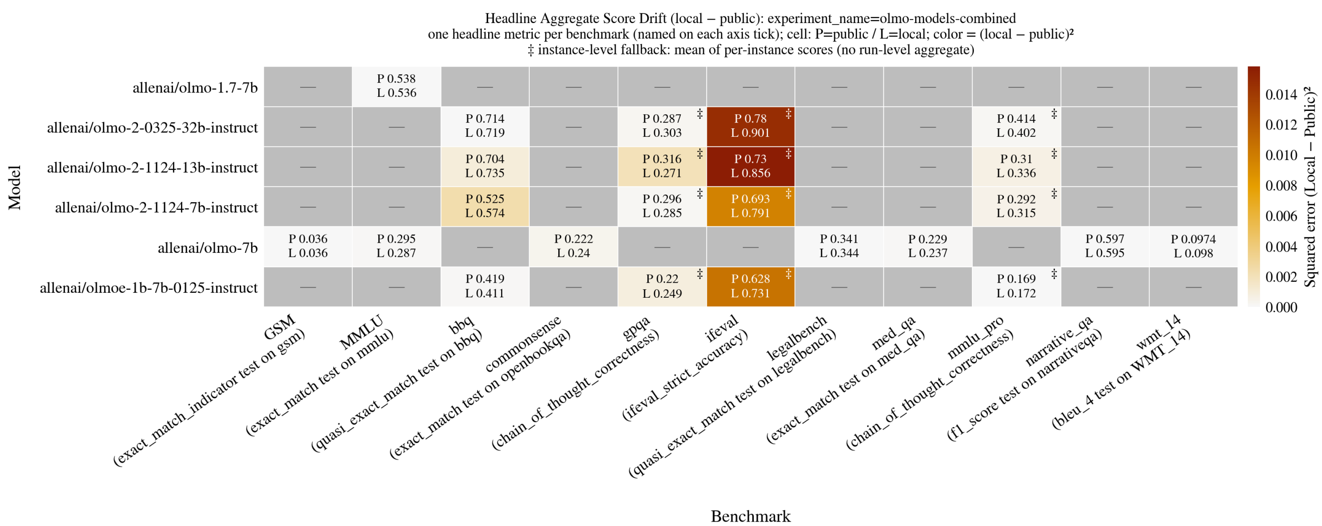
Per-benchmark agreement *(disk)*:

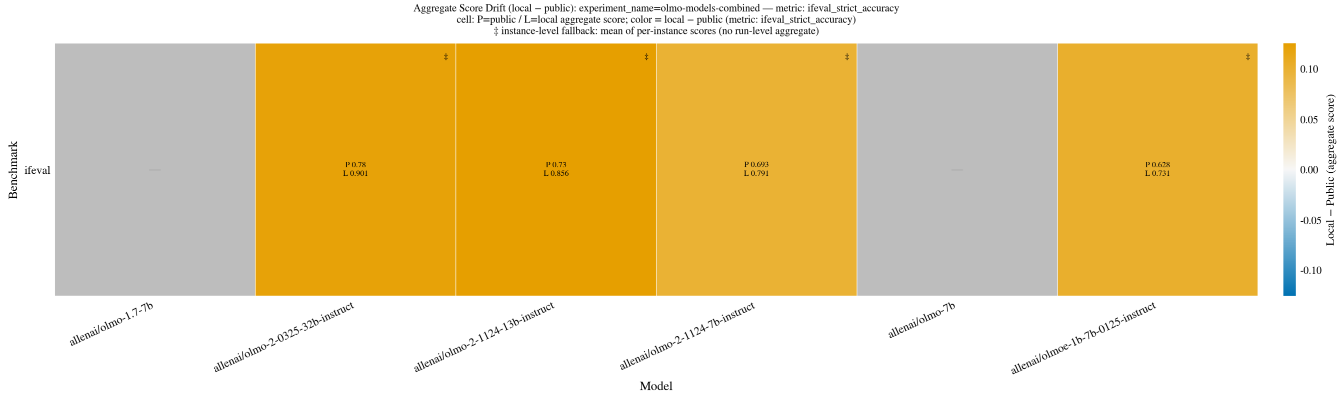
benchmark	runs	mean	median	min
ifeval	4	0.728	0.725	0.673
gpqa	4	0.732	0.712	0.688
mmlu_pro	4	0.821	0.816	0.802
wmt_14	5	0.870	0.882	0.799
commonsense	1	0.926	—	—
med_qa	1	0.936	—	—
bbq	4	0.942	0.936	0.927
narrative_qa	1	0.948	—	—
mmlu	119	0.954	0.965	0.845
legalbench	5	0.973	0.962	0.949
gsm	1	0.990	—	—

Aggregate score drift, official vs local *(disk)* — mean |diff| **0.0312**, median 0.0138, max 0.1257 over 24 headline cells:

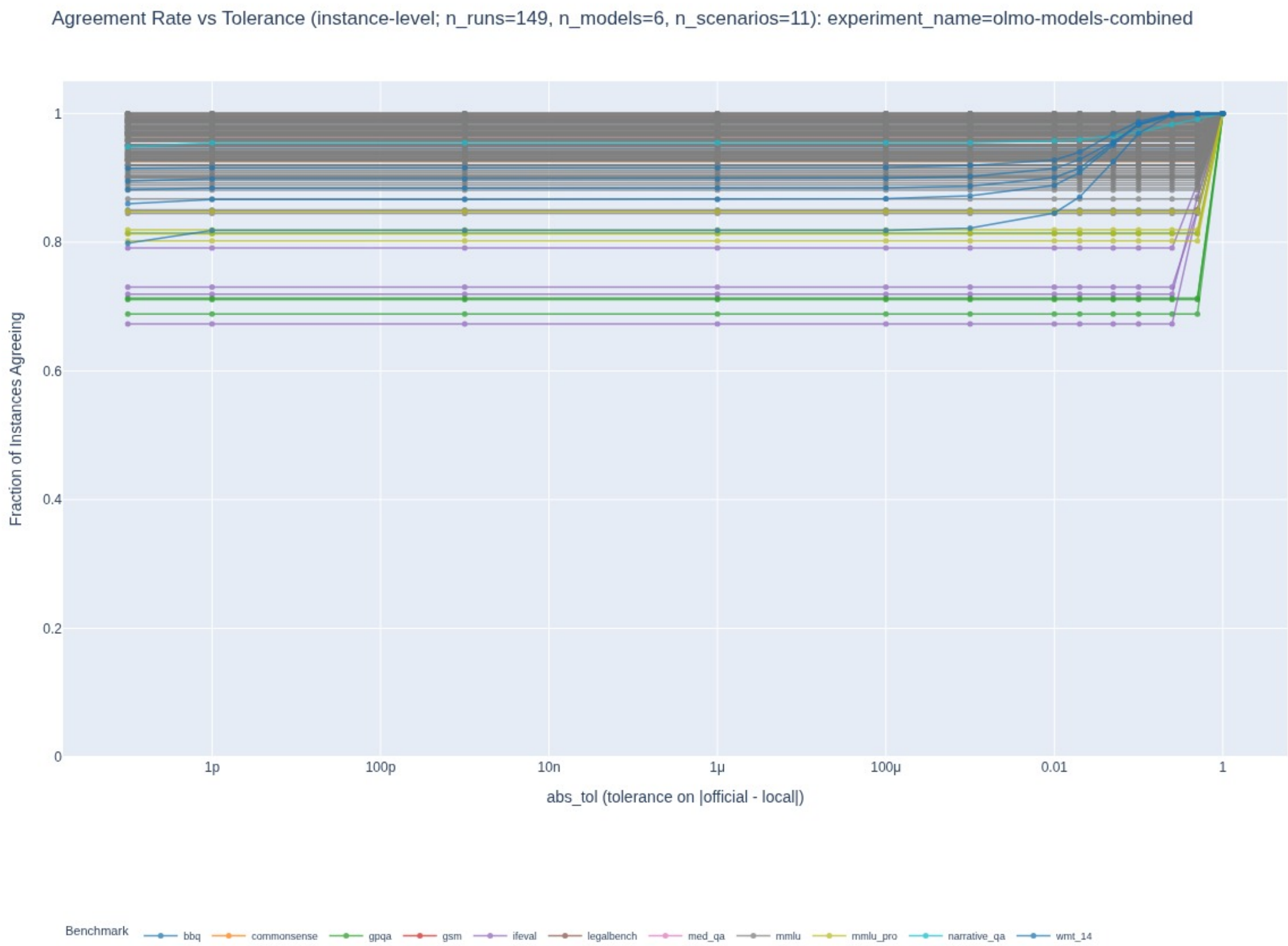
benchmark	cells	mean abs diff	max
ifeval	4	**0.1121**	0.1257
gpqa	4	0.0252	0.0448
bbq	4	0.0232	0.0490
commonsense	1	0.0180	—
mmlu_pro	4	0.0160	0.0260
med_qa	1	0.0080	—
mmlu	2	0.0052	0.0085
legalbench	1	0.0034	—
narrative_qa	1	0.0023	—
wmt_14	1	0.0006	—
gsm	1	0.0000	—

The four worst cells are all ifeval, all same-signed (local **above** official): 13B 0.7298->0.8555 (+0.1257) · 32B 0.7797->0.9014 (+0.1217) · OLMoE 0.6285->0.7314 (+0.1029) · 7B 0.6929->0.7911 (+0.0983). **That single cell is what experiment A8 resolves.**



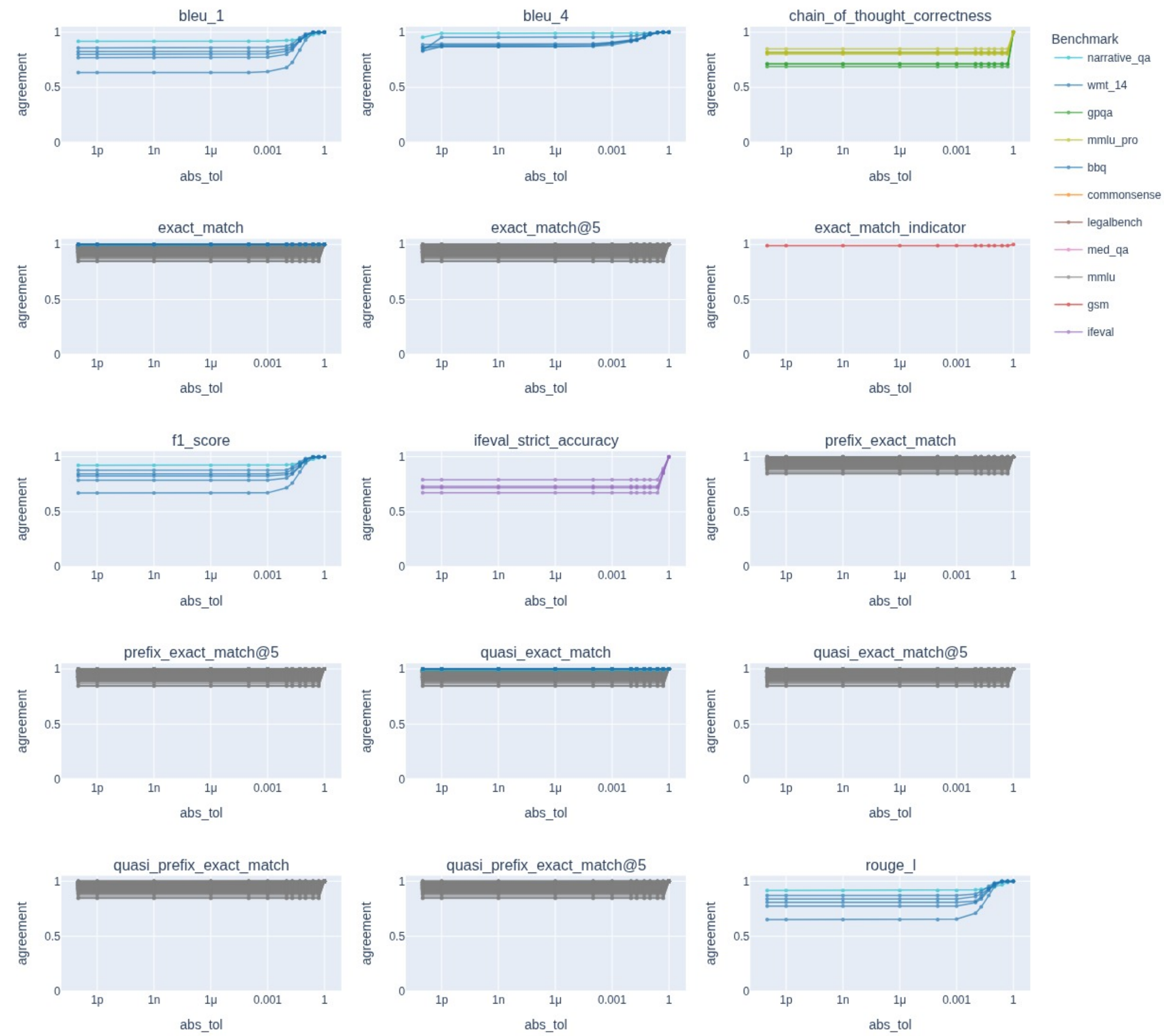


Aggregate Score Drift (local − public): experiment_name=olmo-models-combined — metric: ifeval_strict_accuracy
cell: P=public / L=local aggregate score; color = local − public (metric: ifeval_strict_accuracy)
‡ instance-level fallback: mean of per-instance scores (no run-level aggregate)
Benchmark
ifeval
—
P 0.78
L 0.901
P 0.73
L 0.856
P 0.693
L 0.791
—
P 0.628
L 0.731
allenai/olmo-1.7-7b
allenai/olmo-2-0325-32b-instruct
allenai/olmo-2-1124-13b-instruct
allenai/olmo-2-1124-7b-instruct
allenai/olmo-7b
allenai/olmoe-1b-7b-0125-instruct
Model
Local − Public (aggregate score)
0.10
0.05
0.00
-0.05
-0.10

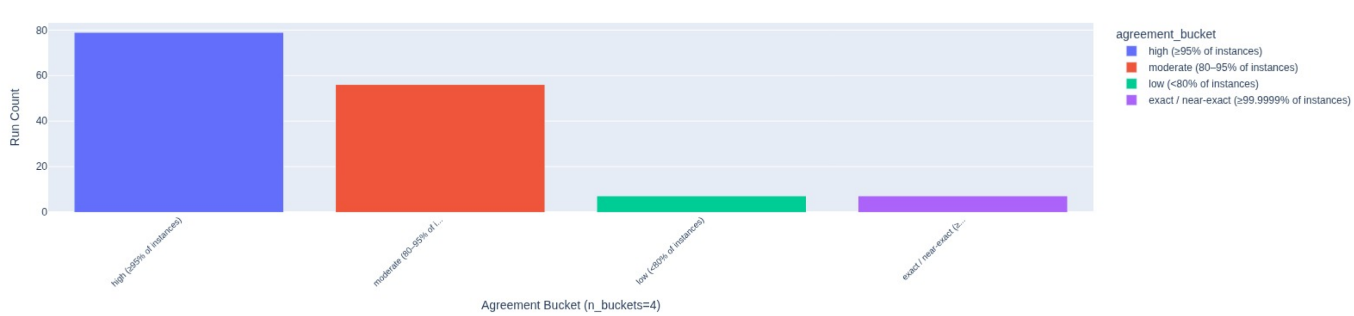


Agreement Rate vs Tolerance (instance-level; n_runs=149, n_models=6, n_scenarios=11): experiment_name=olmo-models-combined
Fraction of Instances Agreeing
1
0.8
0.6
0.4
0.2
0
1p
100p
10n
1μ
100μ
0.01
1
abs_tol (tolerance on |official - local|)
Benchmark
bbq
commonsense
gpqa
gsm
ifeval
legalbench
med_qa
mmlu
mmlu_pro
narrative_qa
wmt_14

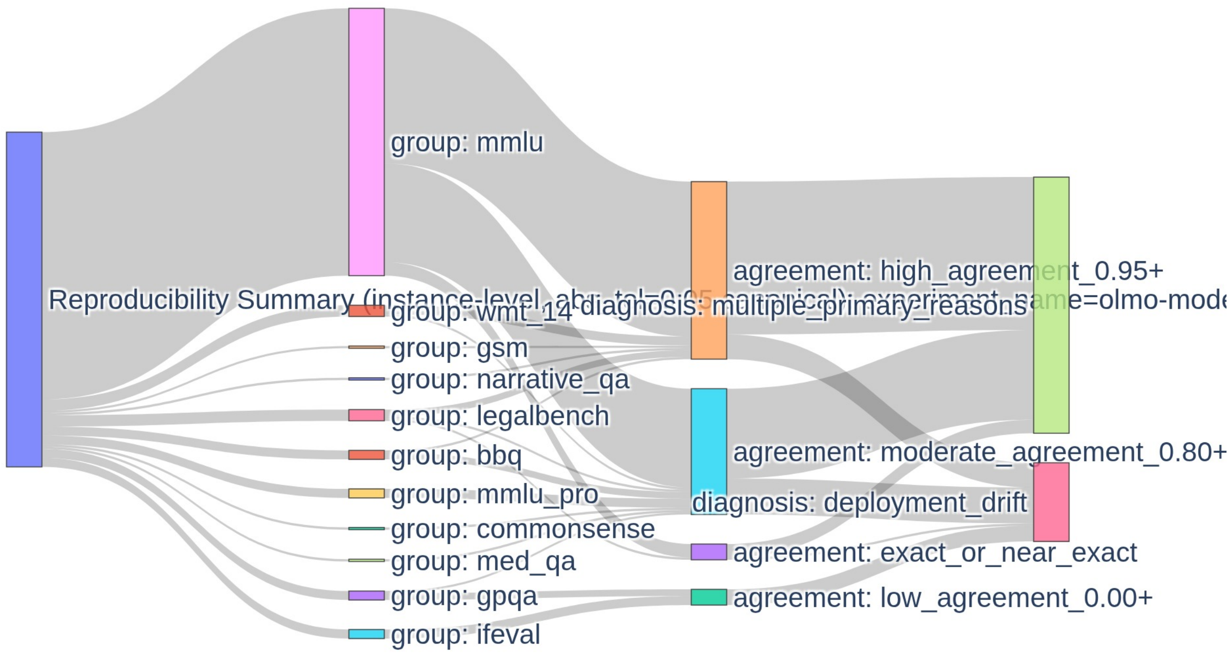
Agreement Rate vs Tolerance (per-metric): experiment_name=olmo-models-combined



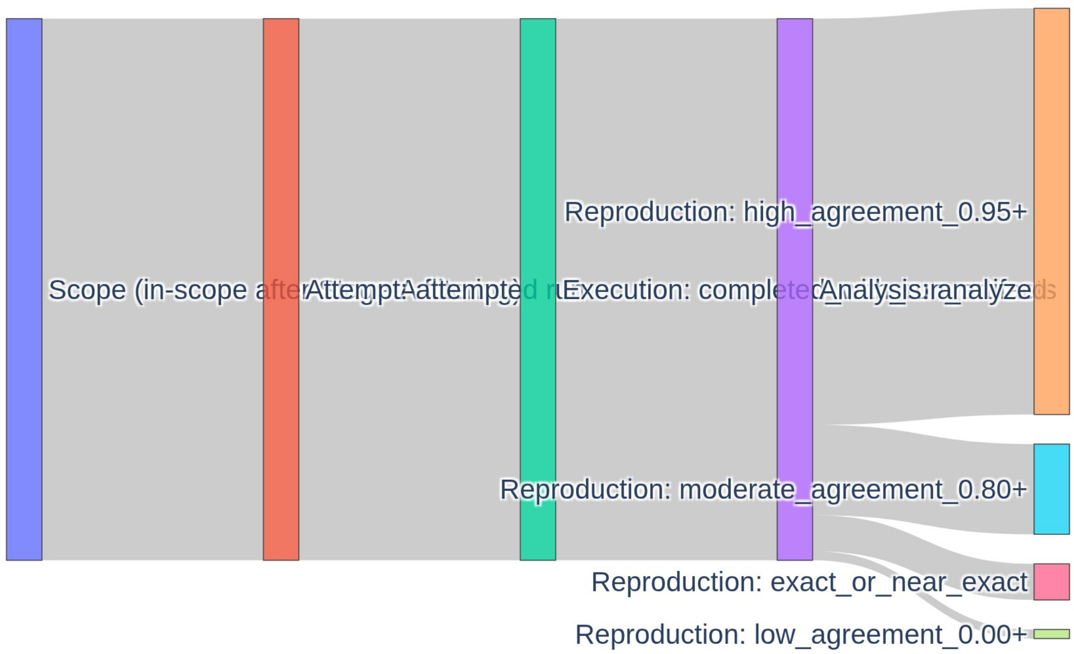
Official vs Local Agreement Buckets (instance-level, abs_tol=0.05 canonical): experiment_name=olmo-models-combined



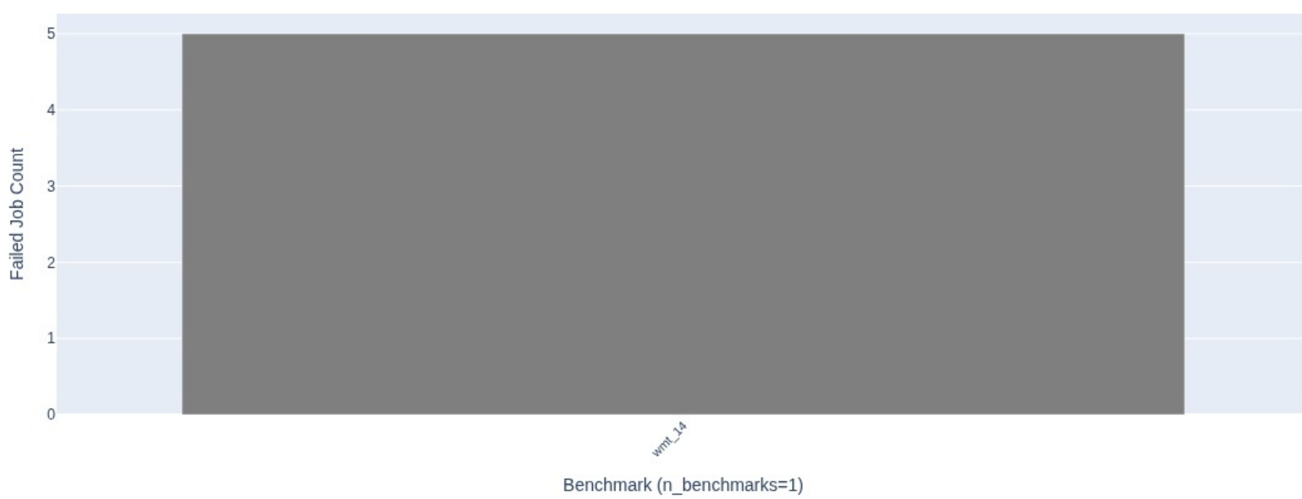
Reproducibility Summary (instance-level, abs_tol=0.05 canonical): experim



Stage B — Scope → Analyzed at abs_tol=0: experiment_name=olmo-mode



Why Jobs Failed: Root Cause Taxonomy by Benchmark: experiment_name=olmo-models-combined



Named causes found and fixed during this campaign (decks 06-23, 06-30, 07-08): OLMo-7B prompt-independent gibberish -> tokenizer appending `<|endoftext|>`; BBQ prompt mismatch -> official carries `output_format_instructions=mcqa`, i.e. genuine recipe drift, not a bug; gpqa gated dataset -> HF auth not reaching the container; ifeval `langdetect` missing -> container needs `crfm-helm[all]`; wmt_14 -> wrong `hf_hub` version.

A4. Qwen reproduction — *the largest and cleanest result*

What. Eight public Qwen text models replayed from frozen `run_spec.json`. **Artifacts.** `virtual-experiments/qwen-models-combined` (report **2026-07-20**); `/data/crfm-helm-audit/audit-qwen-combined-full` — **703 runs with full raw artifacts on disk**.

Coverage *(disk)*. 775 jobs -> 703 with artifacts -> **703 analyzed (100 % of completed)**. 72 failures: 56 `missing_math_dataset`, 16 `unknown_failure`.

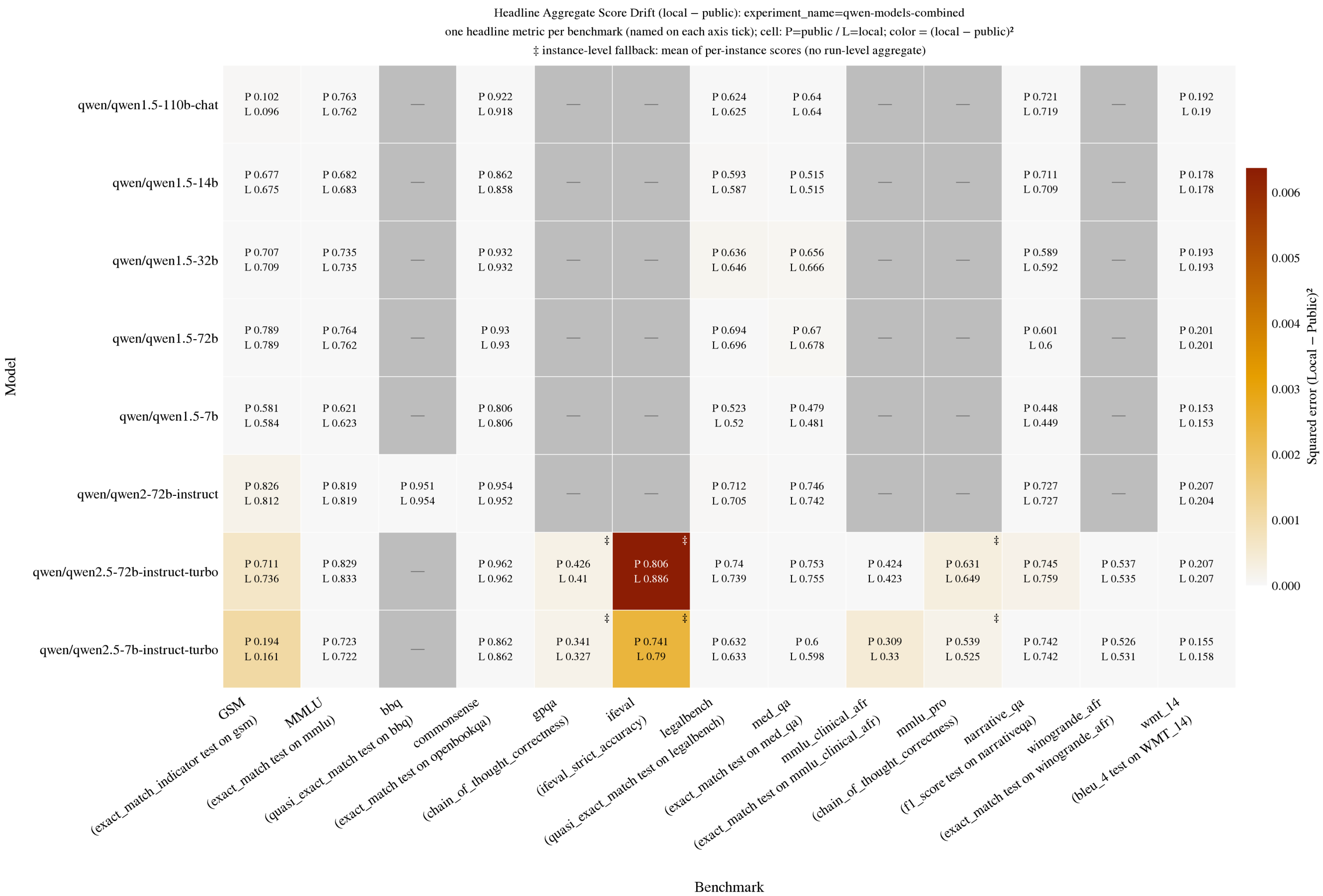
Buckets: high (>=0.95) **474** · exact **159** · moderate 65 · low 5.

Agreement at abs_tol=0 *(disk, 703 runs)*: mean **0.975**, median 0.990, min 0.660, **159 runs at exactly 1.000**.

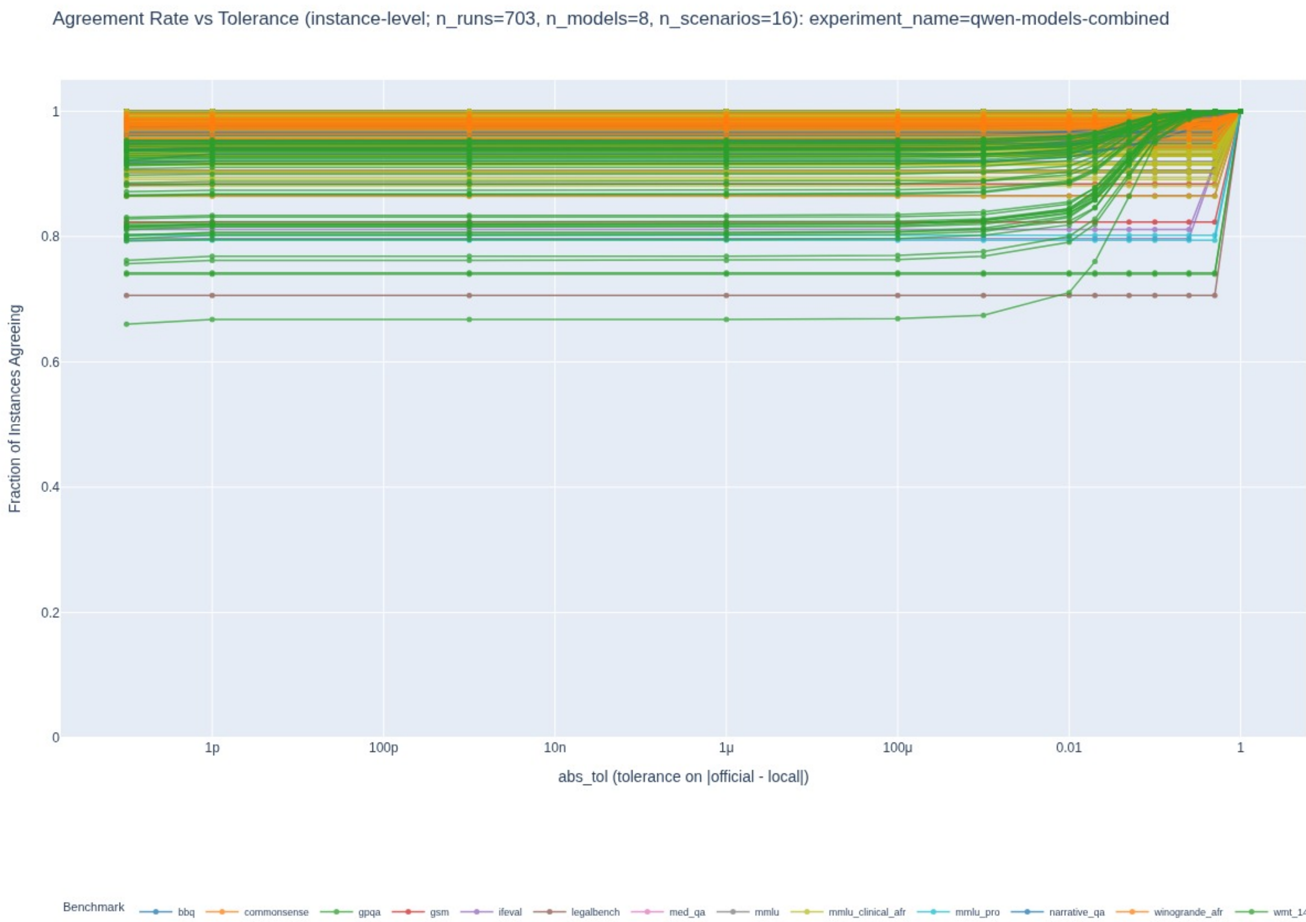
Per-benchmark agreement *(disk)*:

benchmark	runs	mean	median	min	n at 1.000
gpqa	2	0.741	0.741	0.740	0
mmlu_pro	2	0.798	0.798	0.794	0
ifeval	2	0.804	0.804	0.797	0
wmt_14	40	0.873	0.892	0.660	0
gsm	8	0.916	0.947	0.823	0
narrative_qa	8	0.942	0.938	0.917	0
mmlu_clinical_afr	66	0.961	0.980	0.864	3
winogrande_afr	22	0.976	0.979	0.944	0
legalbench	40	0.978	0.993	0.706	15
med_qa	8	0.987	0.988	0.973	0
mmlu	**496**	**0.989**	0.992	0.905	**138**
commonsense	8	0.997	0.997	0.992	3
bbq	1	0.997	—	—	0

Aggregate score drift *(disk)* — mean |diff| **0.0063**, median 0.0020, max 0.0798 over 67 headline cells. **Five times tighter than OLMo.** Same ordering though: ifeval worst (0.0644), then mmlu_pro (0.0160), gpqa (0.0146).

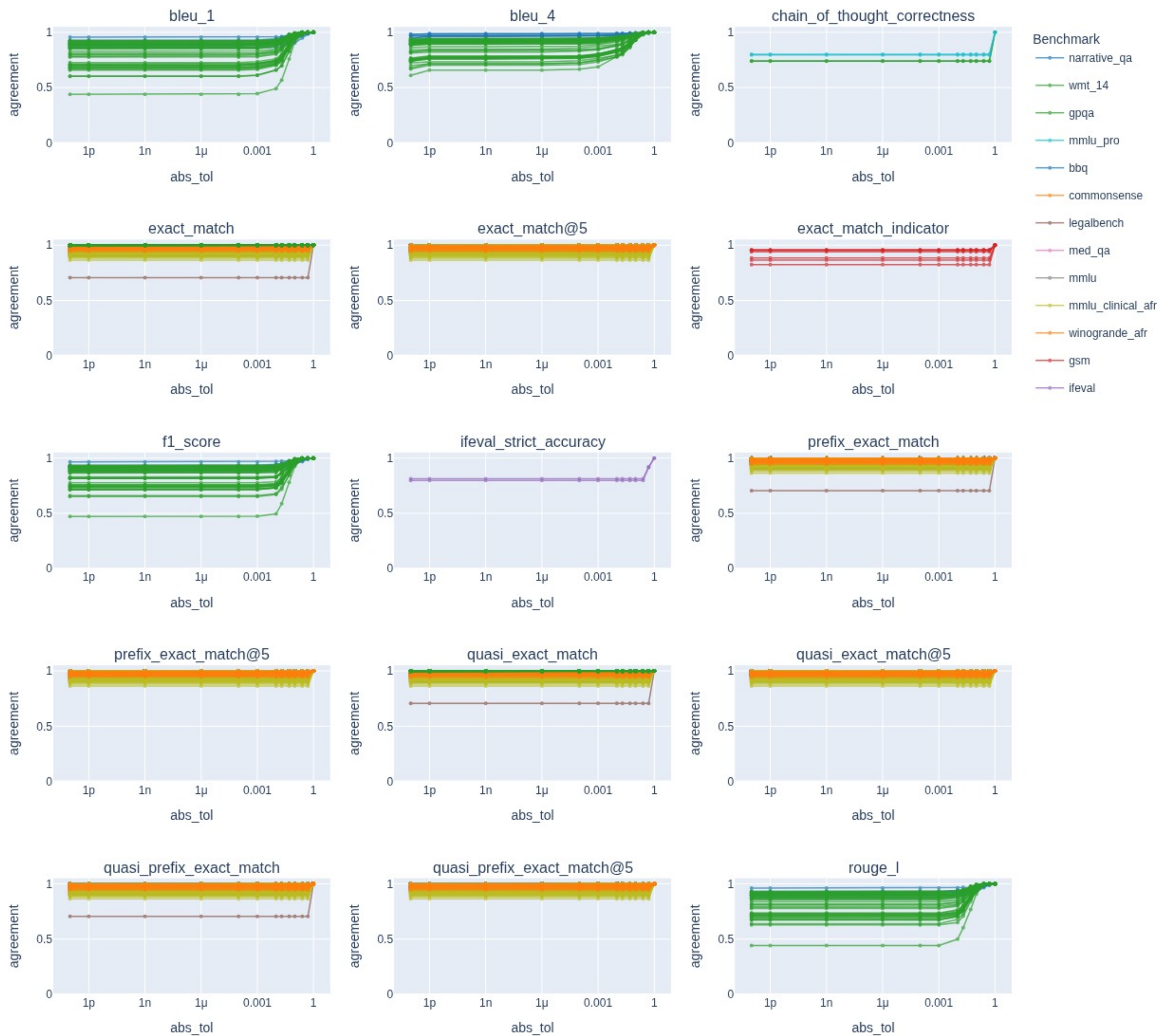


Headline Aggregate Score Drift (local − public): experiment_name=qwen-models-combined
one headline metric per benchmark (named on each axis tick); cell: P=public / L=local; color = (local − public)²
‡ instance-level fallback: mean of per-instance scores (no run-level aggregate)
Model
qwen/qwen1.5-110b-chat
qwen/qwen1.5-14b
qwen/qwen1.5-32b
qwen/qwen1.5-72b
qwen/qwen1.5-7b
qwen/qwen2-72b-instruct
qwen/qwen2.5-72b-instruct-turbo
qwen/qwen2.5-7b-instruct-turbo
P 0.102 L 0.096
P 0.763 L 0.762
P 0.922 L 0.918
P 0.624 L 0.625
P 0.64 L 0.64
P 0.721 L 0.719
P 0.192 L 0.19
P 0.677 L 0.675
P 0.682 L 0.683
P 0.862 L 0.858
P 0.593 L 0.587
P 0.515 L 0.515
P 0.711 L 0.709
P 0.178 L 0.178
P 0.707 L 0.709
P 0.735 L 0.735
P 0.932 L 0.932
P 0.636 L 0.646
P 0.656 L 0.666
P 0.589 L 0.592
P 0.193 L 0.193
P 0.789 L 0.789
P 0.764 L 0.762
P 0.93 L 0.93
P 0.694 L 0.696
P 0.67 L 0.678
P 0.601 L 0.6
P 0.201 L 0.201
P 0.581 L 0.584
P 0.621 L 0.623
P 0.806 L 0.806
P 0.523 L 0.52
P 0.479 L 0.481
P 0.448 L 0.449
P 0.153 L 0.153
P 0.826 L 0.812
P 0.819 L 0.819
P 0.951 L 0.954
P 0.954 L 0.952
P 0.712 L 0.705
P 0.746 L 0.742
P 0.727 L 0.727
P 0.207 L 0.204
P 0.711 L 0.736
P 0.829 L 0.833
P 0.962 L 0.962
‡ P 0.426 L 0.41
‡ P 0.806 L 0.886
P 0.74 L 0.739
P 0.753 L 0.755
P 0.424 L 0.423
‡ P 0.631 L 0.649
P 0.745 L 0.759
P 0.537 L 0.535
P 0.207 L 0.207
P 0.194 L 0.161
P 0.723 L 0.722
P 0.862 L 0.862
‡ P 0.341 L 0.327
‡ P 0.741 L 0.79
P 0.632 L 0.633
P 0.6 L 0.598
P 0.309 L 0.33
‡ P 0.539 L 0.525
P 0.742 L 0.742
P 0.526 L 0.531
P 0.155 L 0.158
GSM (exact_match_indicator test on gsm)
MMLU (exact_match test on mmlu)
bbq (quasi_exact_match test on bbq)
commonsense (exact_match test on openbookqa)
gpqa (chain_of_thought_correctness)
ifeval (ifeval_strict_accuracy)
legalbench (quasi_exact_match test on legalbench)
med_qa (exact_match test on med_qa)
mmlu_clinical_afr (exact_match test on mmlu_clinical_afr)
mmlu_pro (chain_of_thought_correctness)
narrative_qa (f1_score test on narrativeqa)
winogrande_afr (exact_match test on winogrande_afr)
wmt_14 (bleu_4 test on WMT_14)
Benchmark
Squared error (Local − Public)²
0.006
0.005
0.004
0.003
0.002
0.001
0.000

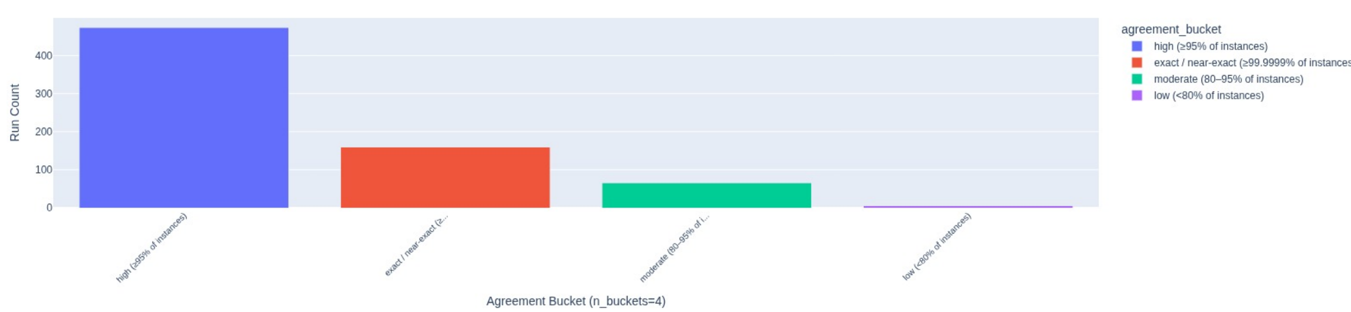


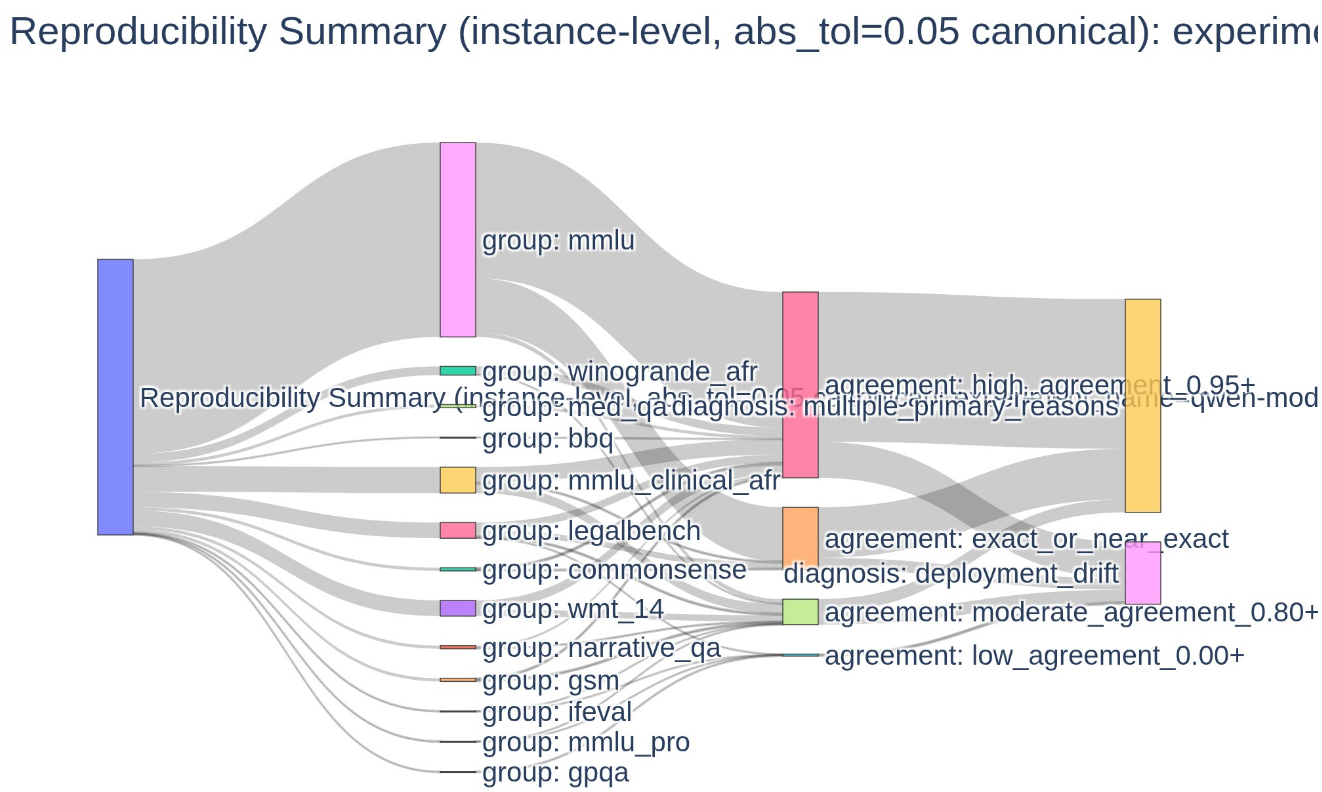
Agreement Rate vs Tolerance (instance-level; n_runs=703, n_models=8, n_scenarios=16): experiment_name=qwen-models-combined
Fraction of Instances Agreeing
1
0.8
0.6
0.4
0.2
0
1p
100p
10n
1µ
100µ
0.01
1
abs_tol (tolerance on |official - local|)
Benchmark
bbq
commonsense
gpqa
gsm
ifeval
legalbench
med_qa
mmlu
mmlu_clinical_afr
mmlu_pro
narrative_qa
winogrande_afr
wmt_14

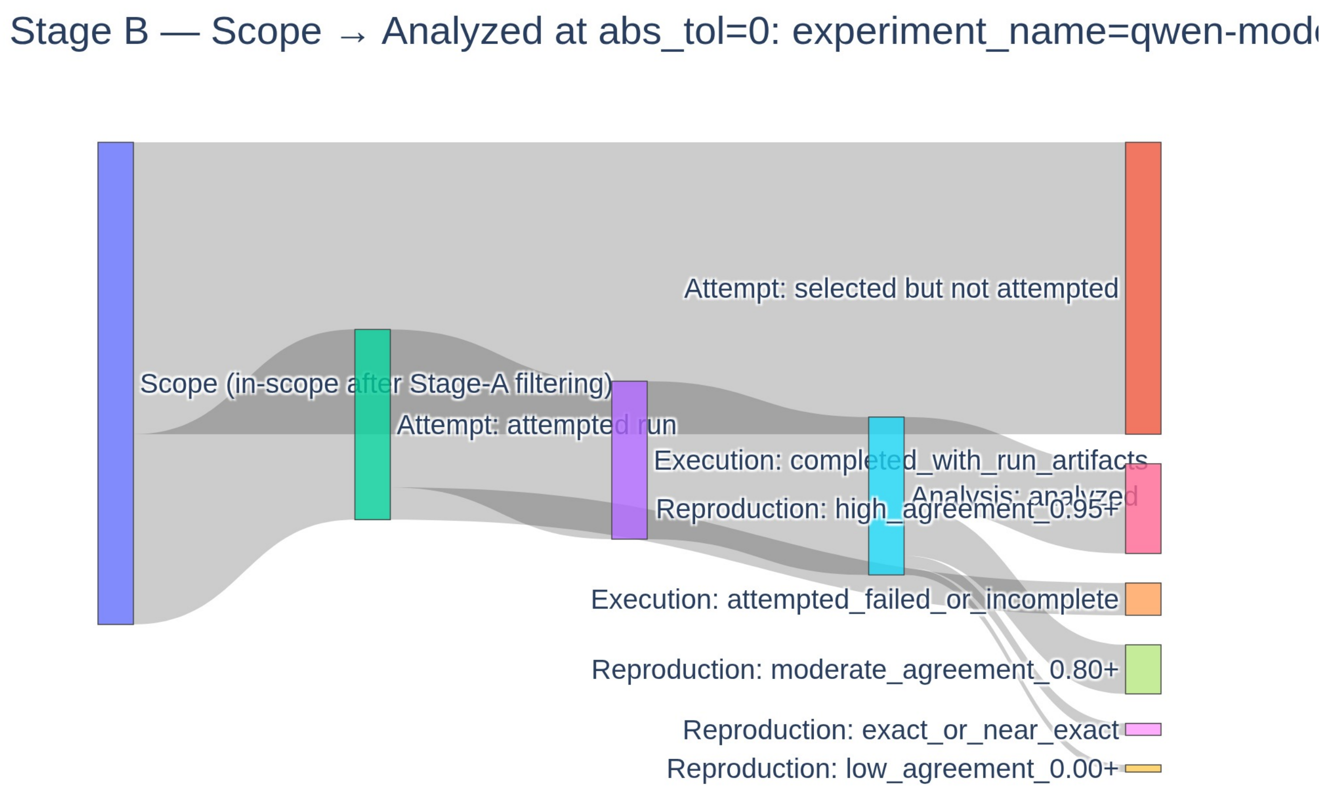
Agreement Rate vs Tolerance (per-metric): experiment_name=qwen-models-combined

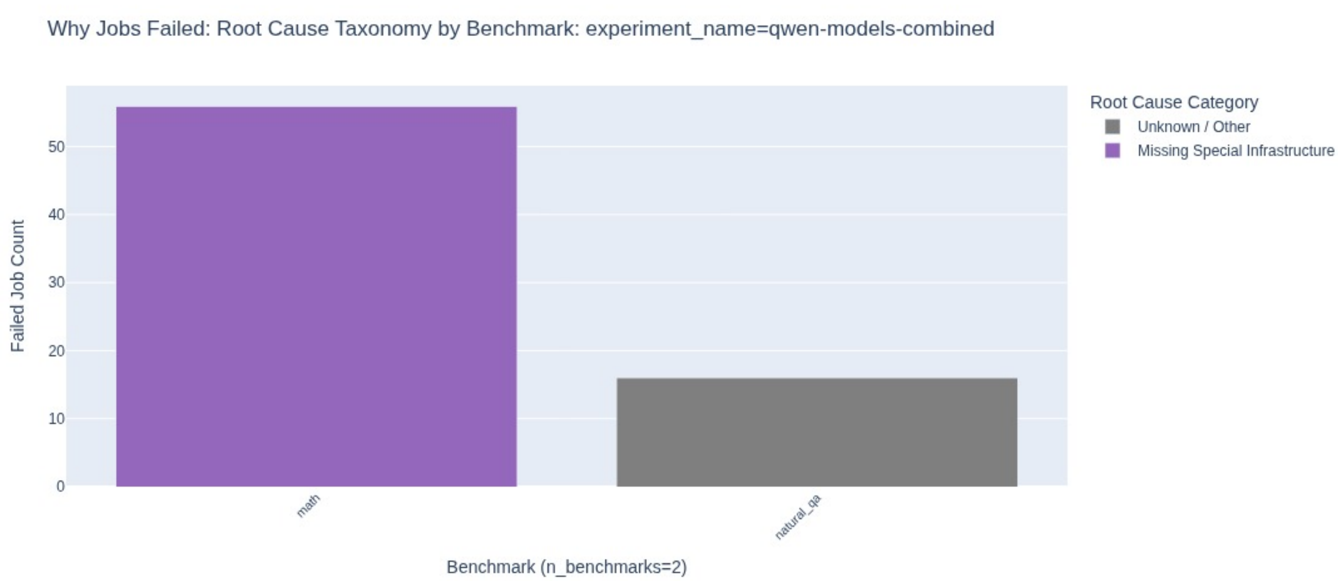


Official vs Local Agreement Buckets (instance-level, abs_tol=0.05 canonical): experiment_name=qwen-models-combined




Reproducibility Summary (instance-level, abs_tol=0.05 canonical): experime
group: mmlu
group: winogrande_afr
group: med_qa
group: bbq
group: mmlu_clinical_afr
group: legalbench
group: commonsense
group: wmt_14
group: narrative_qa
group: gsm
group: ifeval
group: mmlu_pro
group: gpqa
diagnosis: multiple_primary_reasons
diagnosis: deployment_drift
agreement: high_agreement_0.95+
agreement: exact_or_near_exact
agreement: moderate_agreement_0.80+
agreement: low_agreement_0.00+





> **This is the strongest positive reproduction evidence in the project, it is fresh (2026-07-20), and it is barely mentioned in the master reference.** 703 analyzed runs, 159 exact, mean agreement 0.975. It deserves to be a headline, not a footnote. See §8.1.

A5. GPT-OSS-20B from-spec

Artifacts. `virtual-experiments/gpt-oss-20b-from-spec` (report 2026-07-14 17:46 UTC); 4 runs with raw artifacts on disk.

Coverage. 12 jobs -> 4 with artifacts -> 4 analyzed. 8 failures (6 unknown, 2 network/remote-service).

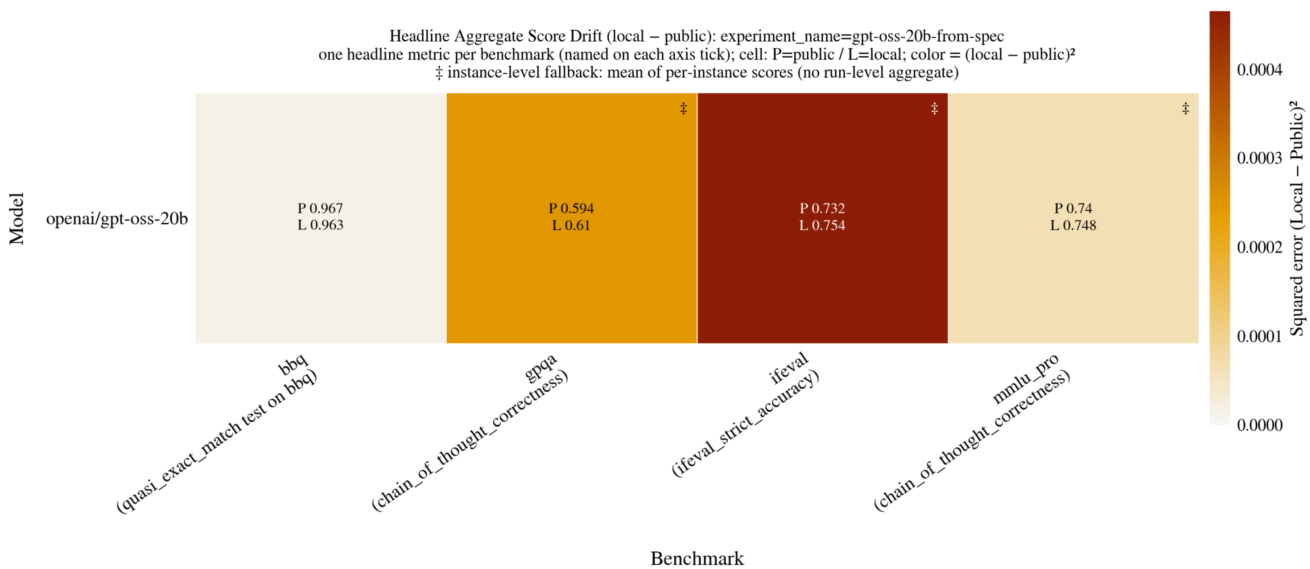
Agreement: mean **0.850**, median 0.855, min 0.712 (n=4).

Score drift *(disk)*, all four cells:

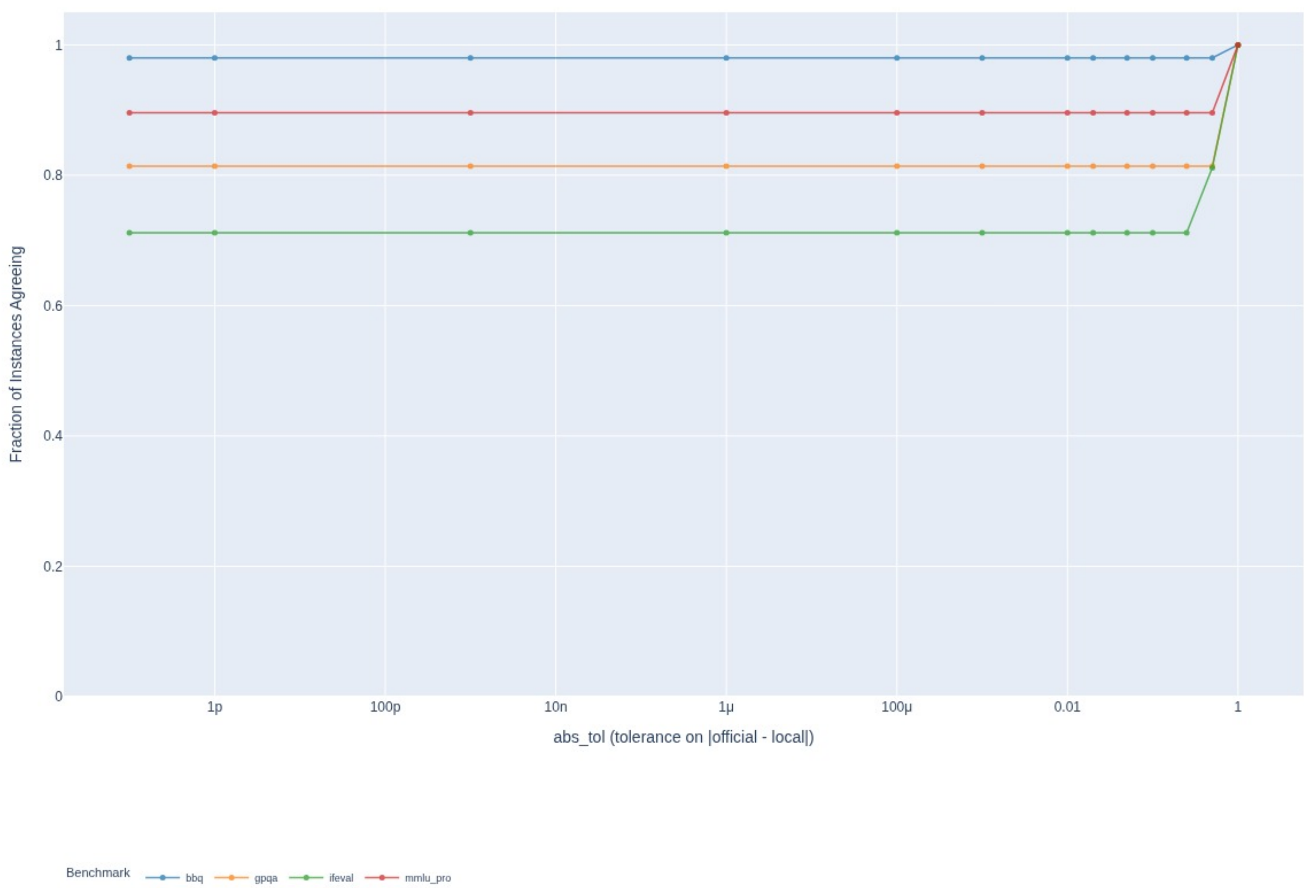
benchmark	official	local	diff
bbq	0.9670	0.9630	–0.0040
mmlu_pro	0.7400	0.7480	+0.0080
gpqa	0.5942	0.6099	+0.0157
ifeval	0.7320	0.7535	+0.0216

Mean |diff| 0.0123; mean squared error 1.98e-04.

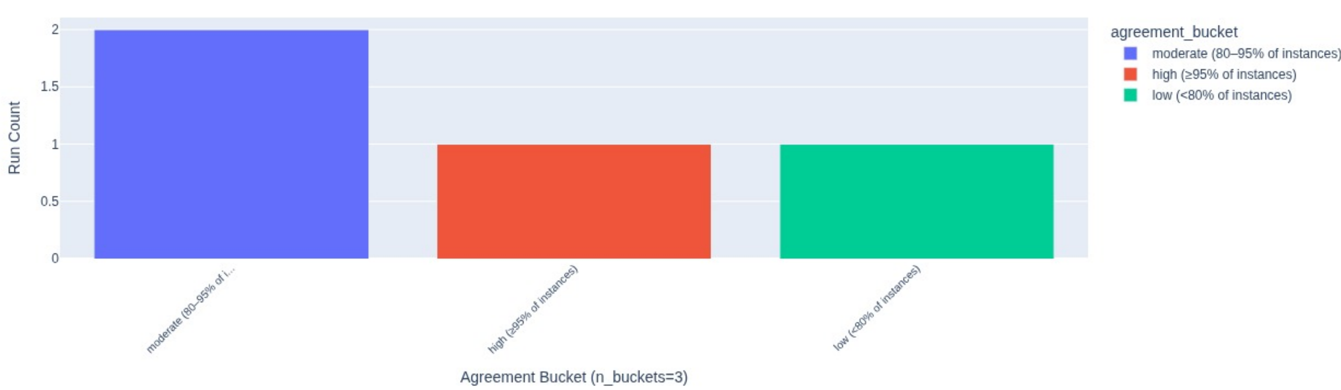
The GPT-OSS-vs-OLMo ifeval ratio, verified today. GPT-OSS ifeval SE = 0.0216² = 4.7e-04. OLMo ifeval SEs = 9.7e-03 ... 1.6e-02. **Ratio 21x–33x** — the master reference's "~30x" is at the top of that range; "25–35x" as the claim ledger states is fair, "two orders of magnitude" (the original wording) is not.

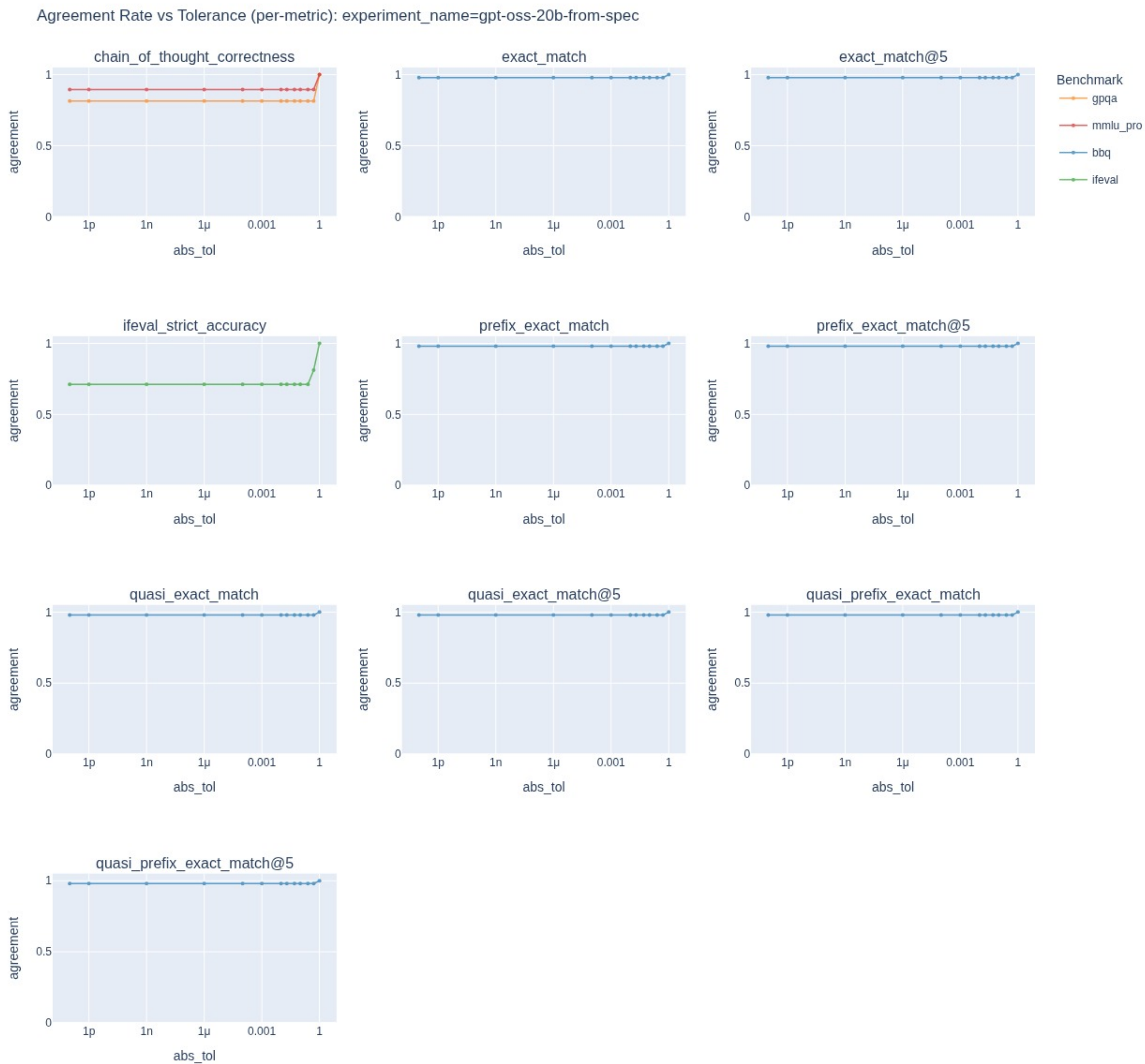


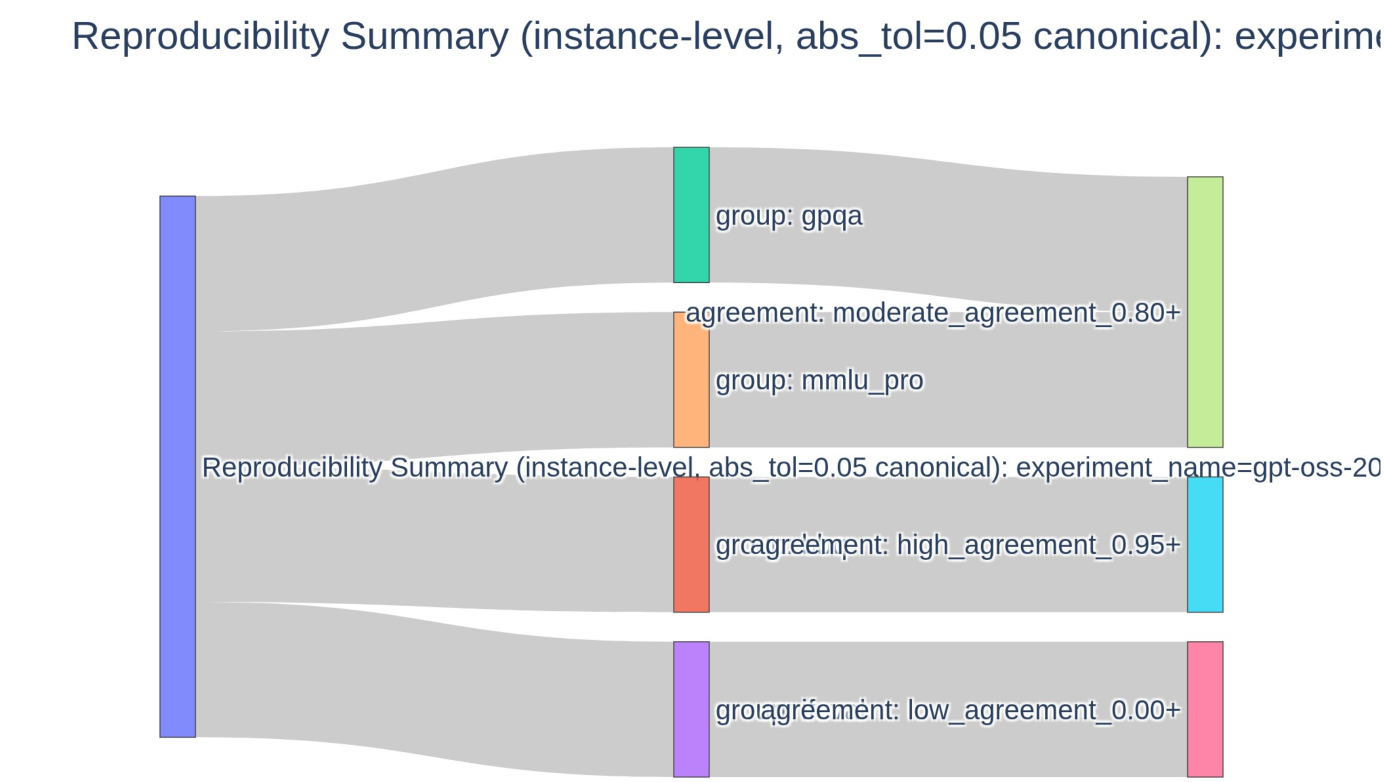
Agreement Rate vs Tolerance (instance-level; n_runs=4, n_models=1, n_scenarios=4): experiment_name=gpt-oss-20b-from-spec

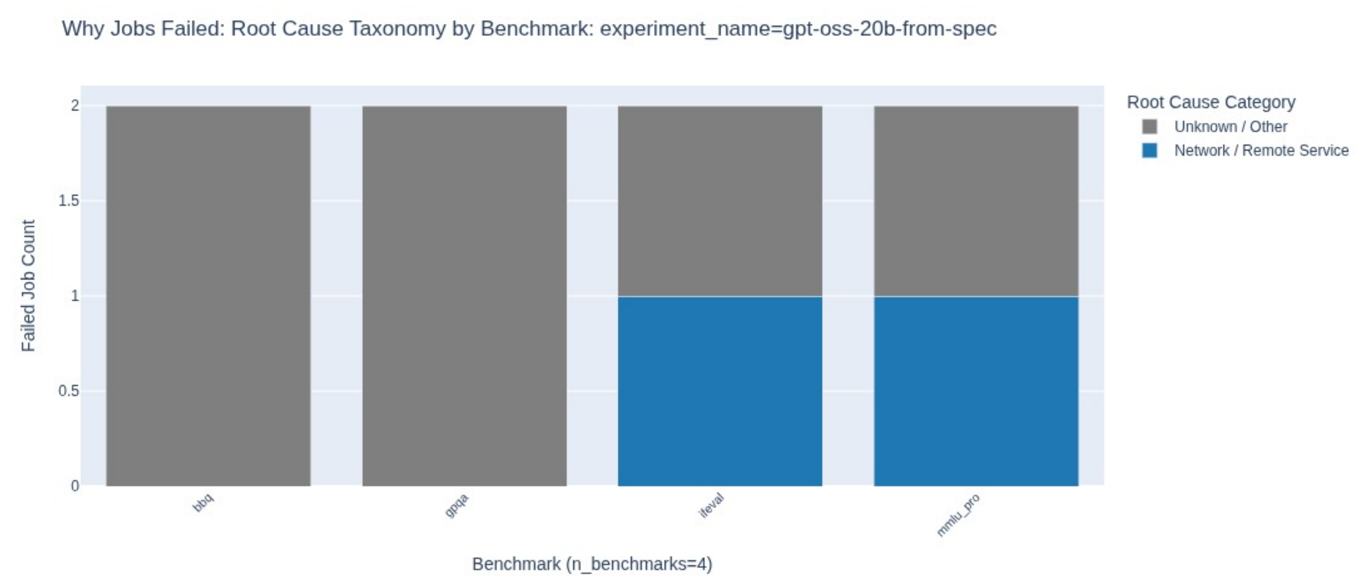


Official vs Local Agreement Buckets (instance-level, abs_tol=0.05 canonical): experiment_name=gpt-oss-20b-from-spec




Agreement Rate vs Tolerance (per-metric): experiment_name=gpt-oss-20b-from-spec
chain_of_thought_correctness
exact_match
exact_match@5
ifeval_strict_accuracy
prefix_exact_match
prefix_exact_match@5
quasi_exact_match
quasi_exact_match@5
quasi_prefix_exact_match
quasi_prefix_exact_match@5
agreement
abs_tol
0
0.5
1
1p
1n
1μ
0.001
1
Benchmark
gpqa
mmlu_pro
bbq
ifeval





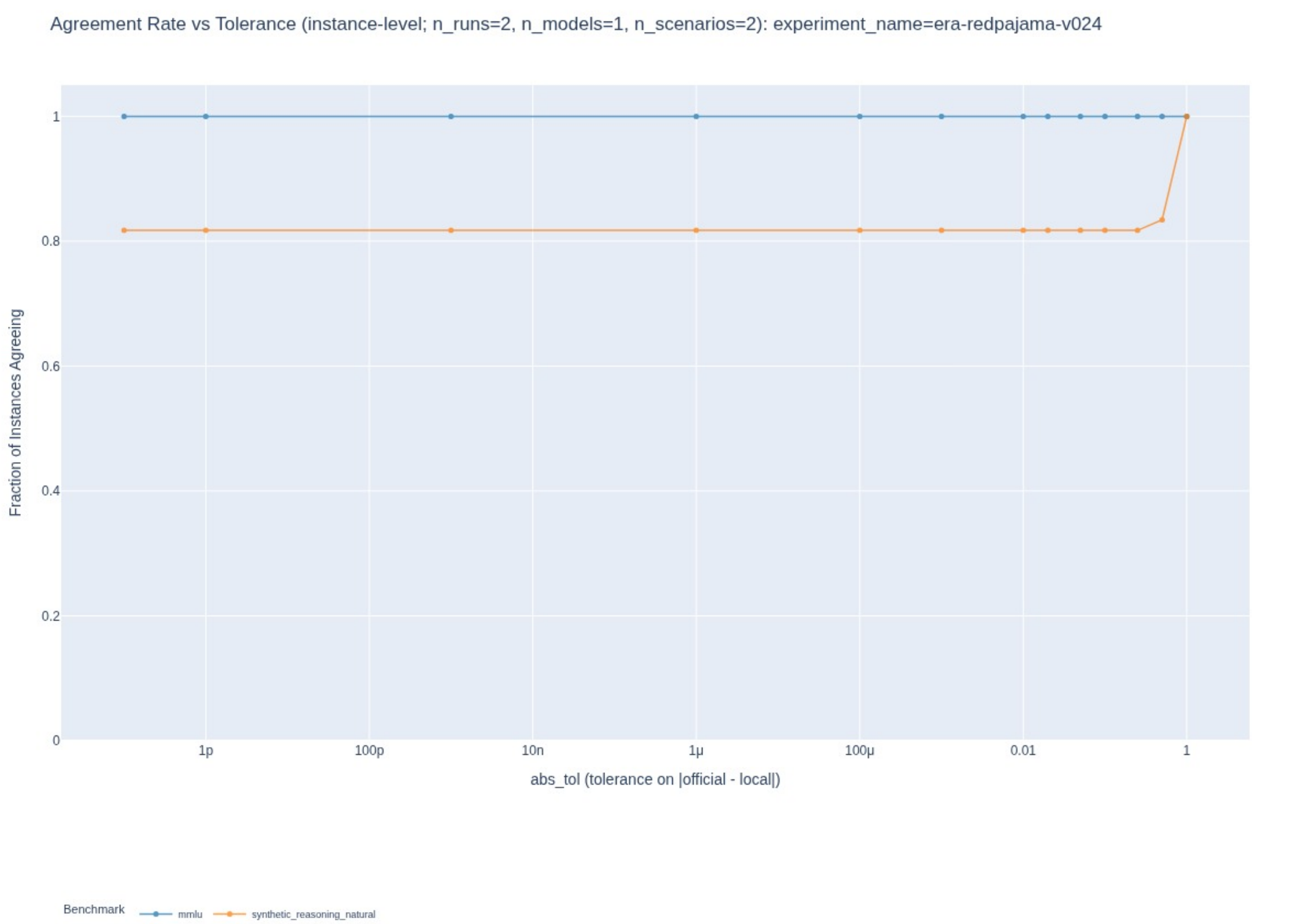
Known engineering cause behind the earlier crash (07-14 deck, resolved): HELM's null-strip crash was a null-vs-`""` serving difference (Together returns `""`, vLLM returns `null`).

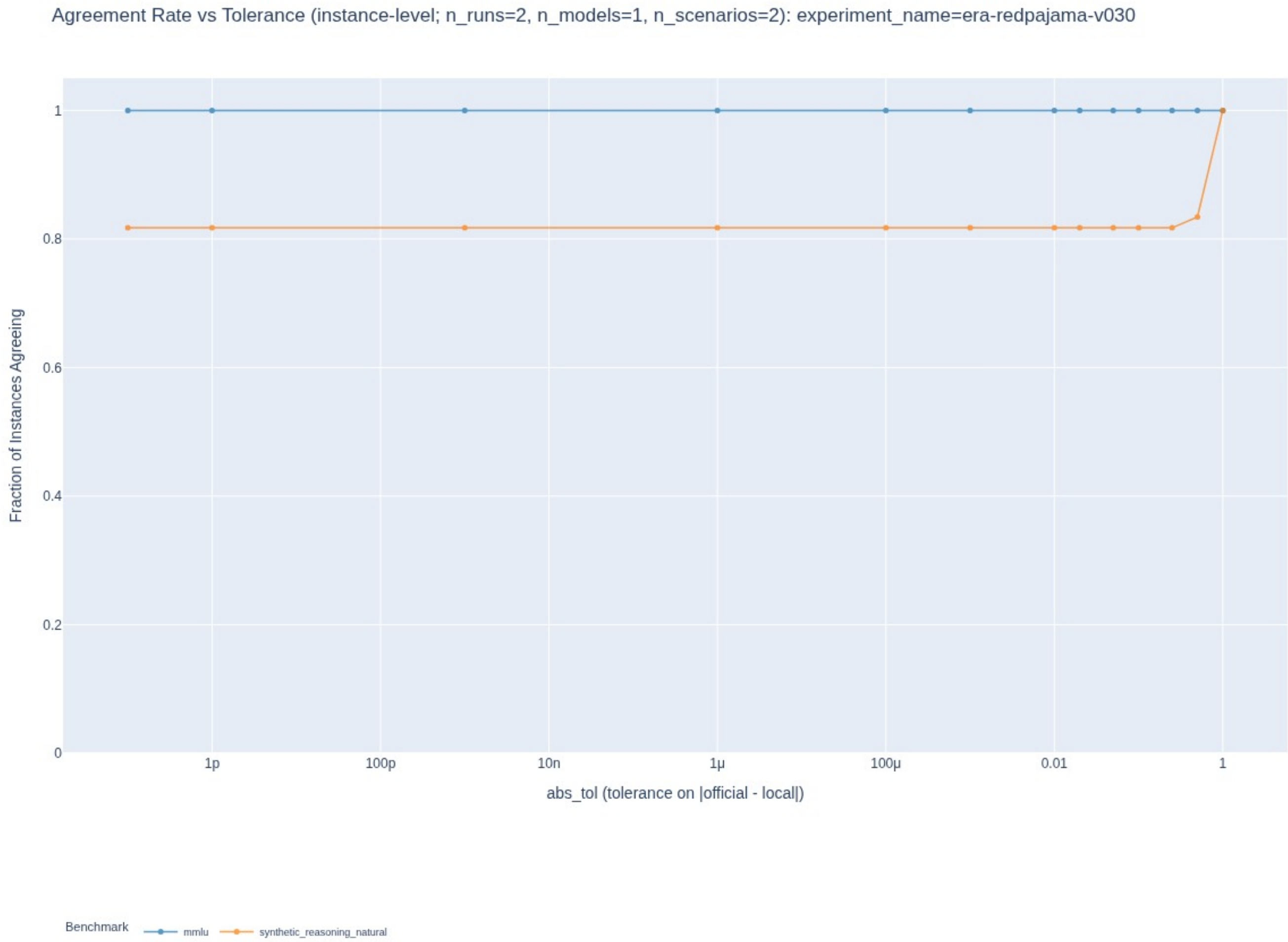
A6. Era-pinned containers — the classic corpus

What. Reproduce pre-v0.5 HELM runs inside version-pinned harness containers, so the *scorer* is a controlled variable. **Artifacts.** `virtual-experiments/era-redpajama-v0{24,30}`; `/data/crfm-helm-audit/era-redpajama_3b-v0_{2_4,3_0}-{full,smoke}` — **16 runs with raw artifacts, present and preservable** (the only stores the 15d audit found intact).

Results *(disk)* — identical under both eras:

era	run	agreement (abs_tol=0)
v0.2.4	`mmlu:subject=us_foreign_policy,...` `redpajama-incite-base-3b-v1`	**1.000**
v0.2.4	`synthetic_reasoning_natural:difficulty=easy,...`	0.817
v0.3.0	`mmlu:subject=us_foreign_policy,...`	**1.000**
v0.3.0	`synthetic_reasoning_natural:difficulty=easy,...`	0.817





The two eras agreeing is not two confirmations. The public artifacts for these runs are byte-identical across v0.2.4 and v0.3.0 — HELM carried them forward rather than re-running. Same public numbers, different local instrument.

G13 (the negative result). GPT-J 6B / GPT-NeoX 20B / OPT 66B run specs name a class path (`helm.benchmark.basic_metrics.BasicMetric`) that resolves in **no** released HELM — a once-migrated hybrid pinning the producing code to unreleased pre-v0.1.0 HELM (2022-07-31 ... 08-26). One `era-gptj_6b-v0_2_4-smoke` directory exists; no runs. *This is a result, not a gap:* the target is underdetermined.

A7. Deployment-match sweeps

What. Serve-and-probe a grid of deployment configurations, score each against the official completions on a sample, rank. **Artifacts.** `/data/crfm-helm-audit-store/deployment-match/` — 26 dirs, 19 with a scored `ranking.txt`, 46+ sweep JSONs.

All scored sweeps, best cell *(disk)*:

sweep	n	cells	best cell	verdict	score	quasi	first-tok
olmoe-1b-7b--ifeval-hf	12	96	fp32 eager agp0	MATCH	0.971	1.00	0.92
olmoe-hf-fp32	12	2	fp32 agp0	MATCH	0.971	1.00	0.92
olmo-2-7b--ifeval-vllm	12	64	fp32 agp0	MATCH	0.915	0.83	1.00

olmo-2-13b--ifeval-vllm	12	64	fp32 agp0	MATCH	0.904	0.83	1.00
olmo-2-32b--ifeval-vllm	12	64	**fp32-tp2** agp0	MATCH	0.961	0.92	1.00
marin-8b--ifeval-chat-vllm	12	64	**fp32 agp0 chat**	MATCH	**1.000**	1.00	1.00
marin-8b--ifeval-vllm	12	32	*(protocol unresolved)*	PARTIAL	0.158	0.00	0.42
olmo-2-7b--ifeval-hf	12	96	fp16/fp32 agp0	PARTIAL	0.324	0.00	0.83
olmo-2-13b--ifeval-hf	12	96	fp16/fp32 agp0	PARTIAL	0.324	0.00	0.83
olmo-2-32b--ifeval-hf	12	48	fp16 agp0	PARTIAL	0.235	0.00	0.58
olmo-2-7b--ifeval-hf-fp32	32	4	fp32 eager agp0 greedy	**MATCH**	**1.000**	1.00	1.00
olmo-2-13b--ifeval-hf-fp32	32	4	fp32 eager agp0 helm	**MATCH**	**1.000**	1.00	1.00
olmo-2-7b--ifeval-hf-fp32-fullN	**541**	16	fp32 eager agp0 helm	**MATCH**	**1.000**	1.00	1.00
olmo-2-13b--ifeval-hf-fp32-fullN	541	—	—	**never ran**	—	—	—

Two findings visible only in this table.

1. **`marin-8b-instruct` — an unreported cross-family confirmation.** fp32 scores 1.000, fp16 0.909, and `auto`/bf16 collapse to 0.517. Marin has an unpinned dtype, so this is the registered "unpinned ⇒ fp32 wins" prediction **confirmed on a second family**. The master reference lists this prediction as untested (it was, at the time — the first marin sweep silently defaulted to the completions protocol and was void; the chat rerun exists and passed). See §8.3.
2. **The old `--ifeval-hf` sweeps are the "residual puzzle," and they are an artifact.** Those 96-cell sweeps rank fp32-agp0 at PARTIAL 0.324; the 4-cell `-hf-fp32` sweep ranks the *same-named cell* at MATCH 1.000. The difference is a bug fix — the probe's `agp`/`ast` arguments were being passed as `0`/`1` where the CLI expected `true`/`false` (fixed 2026-07-23, `3307e000`). The old sweeps were asking for a configuration they did not get.

A8. The fp32 substrate closure — *the flagship*

What. The confirm step the 2026-07-15 consensus demanded, followed by an engine-attribution probe. **Artifacts.** `/data/crfm-helm-audit/audit-allenai-olmo-2-1124-{7b,13b}-instruct-ifeval-fp32` (1 run each, raw artifacts present); the four `hf-fp32` sweeps above.

Step 1 — ordinary-path float32 replay at full n *(disk)*. Denominators verified equal, 1,082 per-instance rows per side per model.

model	official	local fp32+agp0 (vLLM)	D_fp32	prior bf16 gap	closed
OLMo-2-7B-Instruct	0.6929	0.7597	**+0.067**	+0.098	32 %
OLMo-2-13B-Instruct	0.7298	0.8121	**+0.082**	+0.126	35 %

The registered prediction (D_fp32 -> 0) was **refuted**. Directionally right, magnitude wrong. Same closure fraction (~1/3) and same sign on both sizes ⇒ systematic, not noise.

Step 2 — reproduce on the original engine (HF `transformers.generate`). At n=32, all four forward-pass cells MATCH 1.000 on both models.

Step 3 — full-N verification *(disk, recomputed byte-by-byte today)*.

cell (7B, fp32, agp0, decode=helm)	n	byte-identical	ratio
eager, device_map=auto	541	**539**	**0.9963**
eager, device_map=single	541	**539**	**0.9963**
eager, agp**1** (modern template)	541	13	0.0240
sdpa (all variants)	20	20	1.0000 *(incomplete cell)*

> [!] **The probe's own `exact` score reports 1.000; direct byte comparison gives 539/541.** The two non-matching instances (`id13`, `id337`) are long-form generations that diverge mid-stream — at character 153 ("his partner Calvert Vaux" -> "his associate Calvert Vaux") and character 189. These are single-token near-tie flips that cascade. **Report 539/541 (99.63 %), not 541/541.** See §8.4.

Step 4 — attribution. HF-fp32 with *true greedy* decoding also reproduces the official, and greedy vs HELM's `do_sample=True, temp=1e-7` produce byte-identical completions on all 32 probe instances. Decode semantics are therefore **not** the cause.

The decomposition:

recipe	substrate recovered	ifeval drift vs official
bf16 + modern chat template	none	**+0.10**
vLLM fp32 + old-template rendering	precision, template	**+0.07**
HF fp32 + old-template rendering (either decode)	precision, template, **engine**	**~ exact** (539/541)

Same weights, same float32, same prompt, same greedy decode — only vLLM vs Hugging Face — moves a published benchmark metric by +0.067/+0.082.

Not established: one benchmark, one family, two sizes, instruct-only; completion-level evidence, not a metric-level in-process HELM run; the sign of the residual is characterized, not explained.

A9. Qwen3.5-9B-Base extension — *runs complete, no report built*

What. The first **compute** experiment: a model with no public HELM entry, evaluated on the same 9-group core roster as the Qwen reproductions. **Artifacts.** `/data/crfm-helm-audit/audit-qwen35-9b-base-vllm-full` — **72 runs, 72 `stats.json`, complete, dated 2026-07-16/17.** No virtual experiment exists.

Results, read straight from `stats.json` *(disk — published here for the first time)*:

benchmark	metric	runs	mean	min	max
mmlu (57 subjects)	exact_match	57	**0.8098**	0.5300	0.9637
commonsense	exact_match	1	0.9480	—	—
gsm	exact_match_indicator	1	0.8060	—	—
med_qa	exact_match	1	0.7237	—	—

legalbench	exact_match	5	0.6668	0.4578	0.8184
narrative_qa	f1_score	1	0.2688	—	—
wmt_14	bleu_4	5	0.1988	0.1156	0.2478
boolq	exact_match (valid)	1	0.4760	—	—

Beside the reproduced Qwen1.5-7B baseline — the comparison this grid was built for, assembled here for the first time (Qwen1.5-7B column = *local reproduction* from A4, so both sides are our own instrument):

benchmark	Qwen1.5-7B (reproduced)	Qwen3.5-9B-Base	delta
mmlu	0.623	**0.810**	+0.187
gsm	0.584	**0.806**	+0.222
commonsense	0.806	**0.948**	+0.142
med_qa	0.481	**0.724**	+0.243
legalbench	0.520	**0.667**	+0.147
narrative_qa (f1)	0.449	0.269	−**0.180**
wmt_14 (bleu_4)	0.153	**0.199**	+0.046

> [!] **This table is indicative, not final.** The two sides are computed differently: the Qwen1.5-7B column is HELM's *aggregate* headline metric, while the Qwen3.5 column is my *mean over per-subject runs* (57 for mmlu, 5 for legalbench and wmt_14). Those coincide only if subject sizes are equal, which they are not. Building the virtual experiment (§5 item 1) computes both sides the same way and supersedes this table. The direction and rough magnitude will hold; the third decimal will not. **`narrative_qa` is the one regression, and the next table explains it.**

`<think>`-tag leakage across the whole grid *(disk, computed today; unperturbed instances only)*. A base model emitting reasoning-format tags on a plain completions prompt — pretraining contamination with reasoning-format data, surfacing as a scoring failure:

benchmark	instances	with `<think>`	rate
narrative_qa	470	322	**68.5 %**
boolq	1,000	502	**50.2 %**
mmlu	14,869	0	0.00 %
wmt_14	5,000	0	0.00 %
med_qa	1,000	0	0.00 %
legalbench	2,047	0	0.00 %
gsm	1,000	0	0.00 %
commonsense	500	0	0.00 %

The split is perfectly clean: leakage fires only on the two free-form generation tasks and is exactly zero on every constrained-answer shape, across 24,416 instances. It is not a property of the prompt content either — the boolq perturbation arms (mild_mix, dialect, gender, person_name) leak at the same 52 % overall rate as canonical.

Consequences: boolq's 0.476 and narrative_qa's 0.269 are **upper bounds depressed by a measured format-leakage rate**, not capability measurements — with `max_tokens=5`, a `<think>` prefix destroys the boolq answer outright. There were zero empty completions, so the newline-tolerant client did its job and this is a different, residual failure mode. Two things follow for the extension study: report these two cells with the leakage rate attached, and note that the same recipe run against an *instruct* model would not show it — which makes leakage rate a measurable property of the base/instruct boundary rather than a bug to fix.

> **Gap:** one `eval-audit-build-virtual-experiment` invocation away from a full report with plots. Nobody has run it. The numbers above are hand-extracted.

A10. Early / pre-split experiments

29 experiment-analysis trees under `/data/crfm-helm-audit-store/reports/core-run-analysis/` (generated 2026-05-20 and 2026-05/06). Most predate the repository split and the from-spec refactor.

experiment	run entries	reports built	note
`audit-historic-grid`	286	**56**	gpt2 (44), sea-lion (10), granite, + 1
`audit-qwen25-7b-aiq`	137	**65**	qwen2.5-7b-instruct-turbo across 10 families
`audit-gpt-oss-20b-vllm-smoke`	2	2	
`audit-small-models-kubeai-overnight`	8	0	all skipped
pythia/vicuna r1/r2 pairs (boolq, mmlu-usfp, narrative)	1 each	0	repeatability pairs, reports never built
`audit-vicuna-nochat-{overnight,server}`	3 each	0	

Also on disk with raw artifacts but **no analysis at all**: `audit-falcon-7b-helm-grid` (41 runs, 2026-04/05) and `audit-gpt-oss-20b-core-grid` (40 runs, 2026-05).

These are the repeatability and cross-machine baselines (r1/r2 pairs = same recipe twice). They were run, the artifacts survive, and the reports were never built. Cheap to recover.

---

3. Thread B — open-weight judges

B1. Identity-replay gate (artifact reconstruction)

What. Reattach the *original* judge annotations to a reconstructed scenario state, run the *real* official metric, and require the published statistics back to 1e-12. Pure reconstruction — no model execution. **Artifact.** `/data/crfm-helm-audit-store/open-judge/replay-report.json`.

Results *(disk)* — 6/6, zero failures:

benchmark	max abs error	aggregate rows	instance rows
xstest	**0.000e+00**	15	2,250
omni_math	**0.000e+00**	3	1,000
harm_bench	**0.000e+00**	15	2,000
simple_safety_tests	**0.000e+00**	15	500
anthropic_red_team	**0.000e+00**	15	5,000

wildbench	1.954e-14	6	2,000

Source audit: 6 runs examined, 6 supported, 0 unsupported (candidate `openai/gpt-oss-20b`).

> These are the cheapest, most citable, and most easily preserved artifacts in the entire project — and they are unhashed and unbundled.

B2–B3. Judge substitution and the size sweep

Artifacts. 75 attempt directories, all `DONE`, under `/data/crfm-helm-audit-store/open-judge/results/`. Grid *(disk)*:

benchmark	0.8B	2B	4B	9B	27B	35B-A3B
xstest (450)	3x450	3x450	3x450	3x450	3x450	3x450
wildbench (1000)	3x1000	3x1000	3x1000	3x1000	3x1000	3x1000
anthropic_red_team (1000)	—	3x1000	3x1000	3x1000	—	—
harm_bench (400)	—	3x400	3x400	3x400	—	—
simple_safety_tests (100)	—	3x100	3x100	3x100	—	—
omni_math	—	—	—	—	—	— **never run**

(cells = replicates x instances; 3 replicates at temperature 0 throughout)

Scores and parse rates, computed from the artifacts today *(disk)*:

benchmark	official	judge	score (mean of 3 reps)	sd across reps	parse ok
xstest	0.8717	0.8B	0.8244	0.0022	1.000
		2B	0.8630	0.0013	1.000
		4B	0.9423	0.0079	0.874
		9B	0.8766	0.0053	0.982
		27B	0.8690	0.0014	0.996
		35B-A3B	0.8691	0.0006	0.999
wildbench	7.6363	0.8B	3.9752	0.1037	**0.141**
		2B	7.2537	0.0221	0.589
		4B	6.9103	0.0231	0.685
		9B	6.5236	0.0160	0.919
		27B	6.7040	0.0249	0.841
		35B-A3B	6.8398	0.0179	0.980
anthropic_red_team	0.9965	**2B**	**0.2549**	0.0032	**0.999**
		4B	0.9997	0.0003	0.942

		9B	0.9996	0.0001	0.990
harm_bench	0.9869	2B	0.9604	0.0071	1.000
		4B	0.9861	0.0009	0.838
		9B	0.9897	0.0020	0.934
simple_safety_tests	1.0000	2B	0.6500	0.0000	1.000
		4B	1.0000	0.0000	0.977
		9B	1.0000	0.0000	1.000

Agreement analysis (v1 arms, 2026-07-19 report):

xstest (label metric, n=450):

pair	signed	abs	pearson	within	kappa
qwen3.5-27B vs official GPT	+0.0093	0.0115	0.958	98.4 %	**0.936**
qwen3.6-35B-A3B vs official GPT	+0.0117	0.0117	0.959	98.2 %	**0.928**
qwen3.5-27B vs qwen3.6-35B	–0.0024	0.0043	0.993	98.7 %	0.944
official GPT vs official Llama	–0.0300	0.0300	0.895	96.0 %	***0.829***

Open-vs-closed disagreement is smaller than the closed ensemble's internal disagreement.

wildbench (1–10 rubric, n~962–999):

pair	signed	abs	pearson	spearman	within
qwen3.5-27B vs official GPT	–0.8228	1.2337	0.826	0.656	63.6 %
qwen3.6-35B vs official GPT	–0.7313	1.0188	0.872	0.700	70.5 %
qwen3.5-27B vs qwen3.6-35B	–0.0869	0.6635	0.935	0.879	83.0 %
official GPT vs official Llama	–0.1271	0.6306	0.936	0.687	91.4 %

Replicate behaviour at temperature 0 — xstest: sd 0.0005, 0.2 % of instances change score. wildbench: sd 0.2145, **38.7–42.6 % of instances change**, max range 5–6 points. Judge *text* differs across replicates on 87–96 % of instances either way. Judge non-determinism is universal; **metric fragility is a property of the metric.**

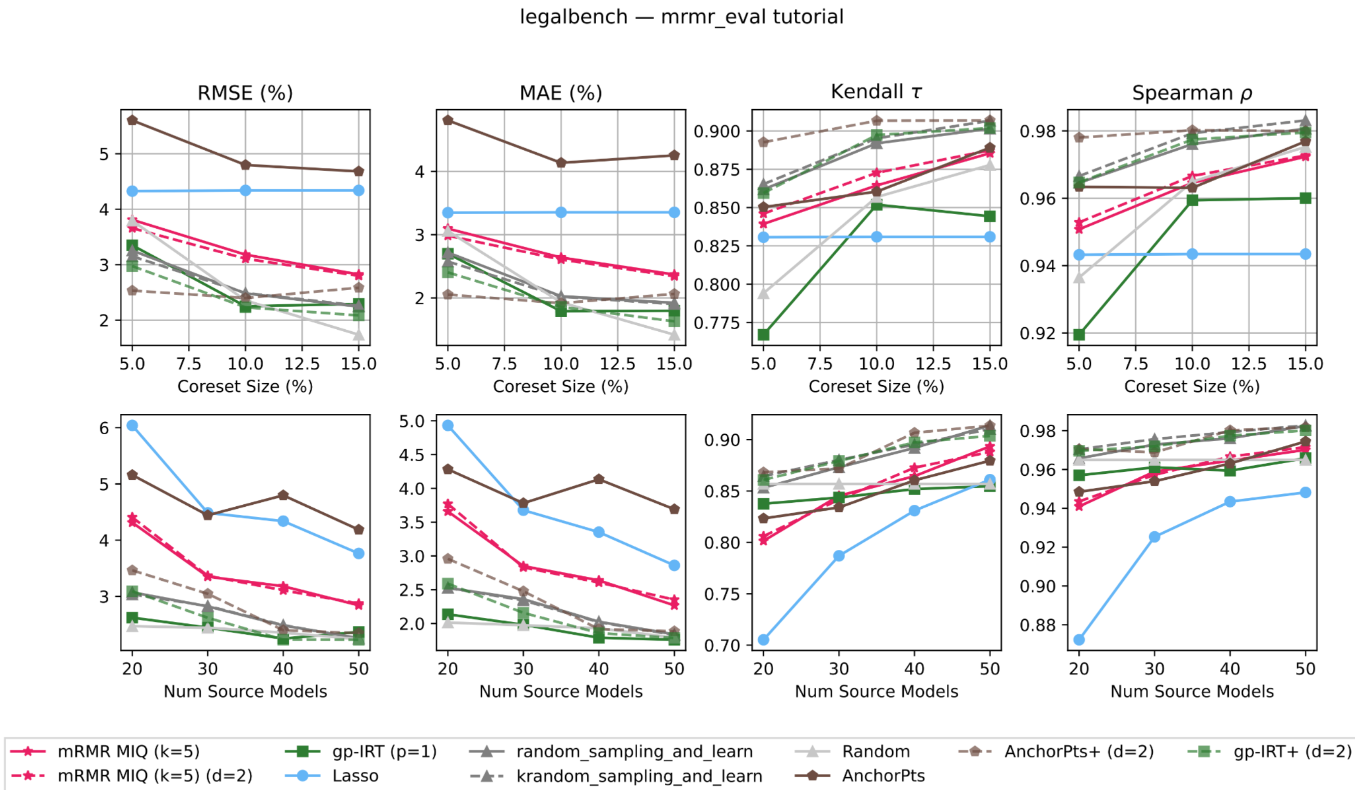
Three caveats that must travel with these numbers. 1. **Single candidate.** Every artifact scores `gpt-oss-20b`. Every endpoint of interest (ranking preservation, conclusion survival) is defined over a *set*. 2. **Parse ≠ calibration.** Qwen3.5-2B: 99.9 % parse, score 0.2549 against an official 0.9965 on red-team — a label inversion that any format-based health check passes. Safety sets are ~99 % one-class, so "agreement" there is essentially a false-positive rate. 3. **Contamination.** Qwen3.5 launched 2026-02-16; the HELM Safety v1.14.0 / Capabilities v1.12.0 publication dates could not be established. Every agreement figure is an **upper bound**.

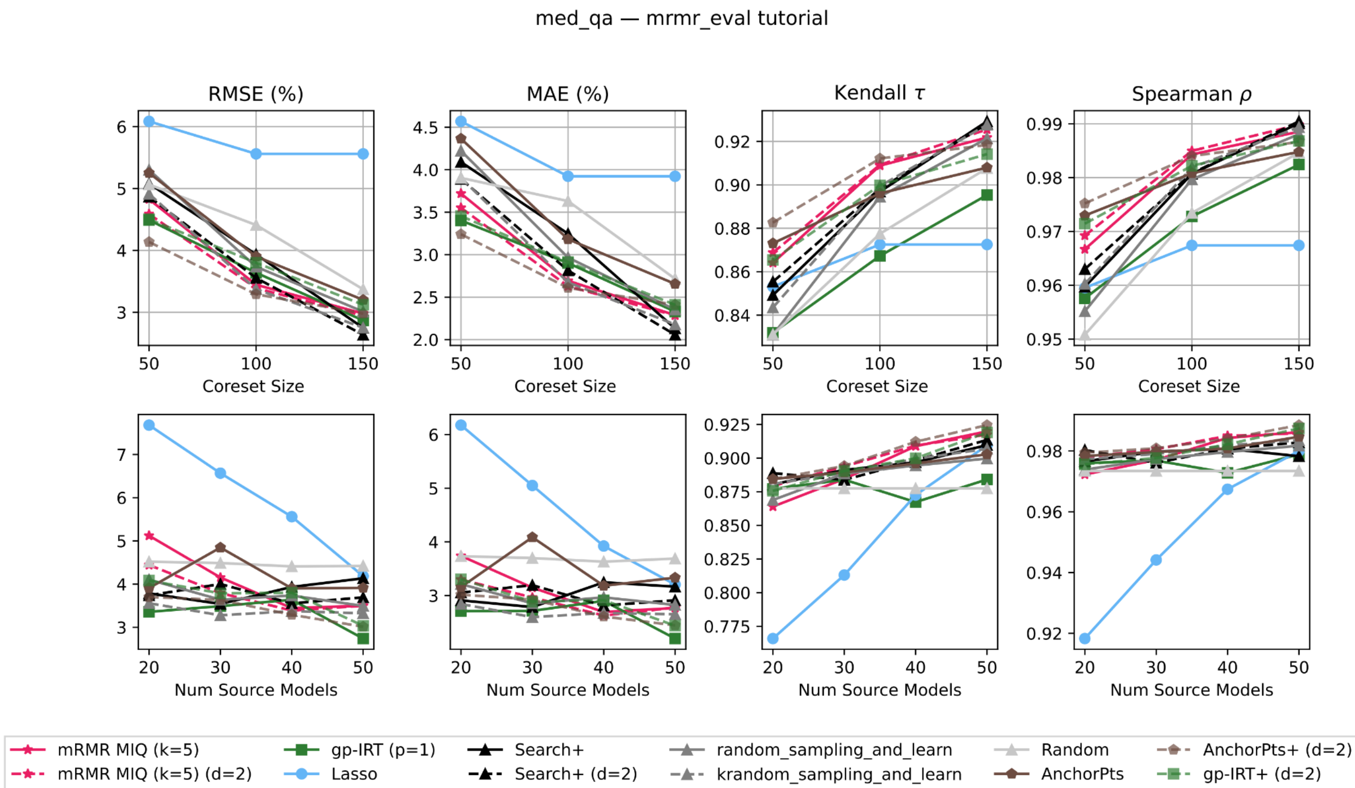
Stale: the only stored analysis reports are the two 2026-07-19 pre-sweep files (27B and 35B-A3B only). The 4 later benchmarks and 4 smaller judges have never been run through `30_analyze_judges.sh`. The table above is my recomputation from raw artifacts; agreement/kappa for those arms does not exist anywhere yet.

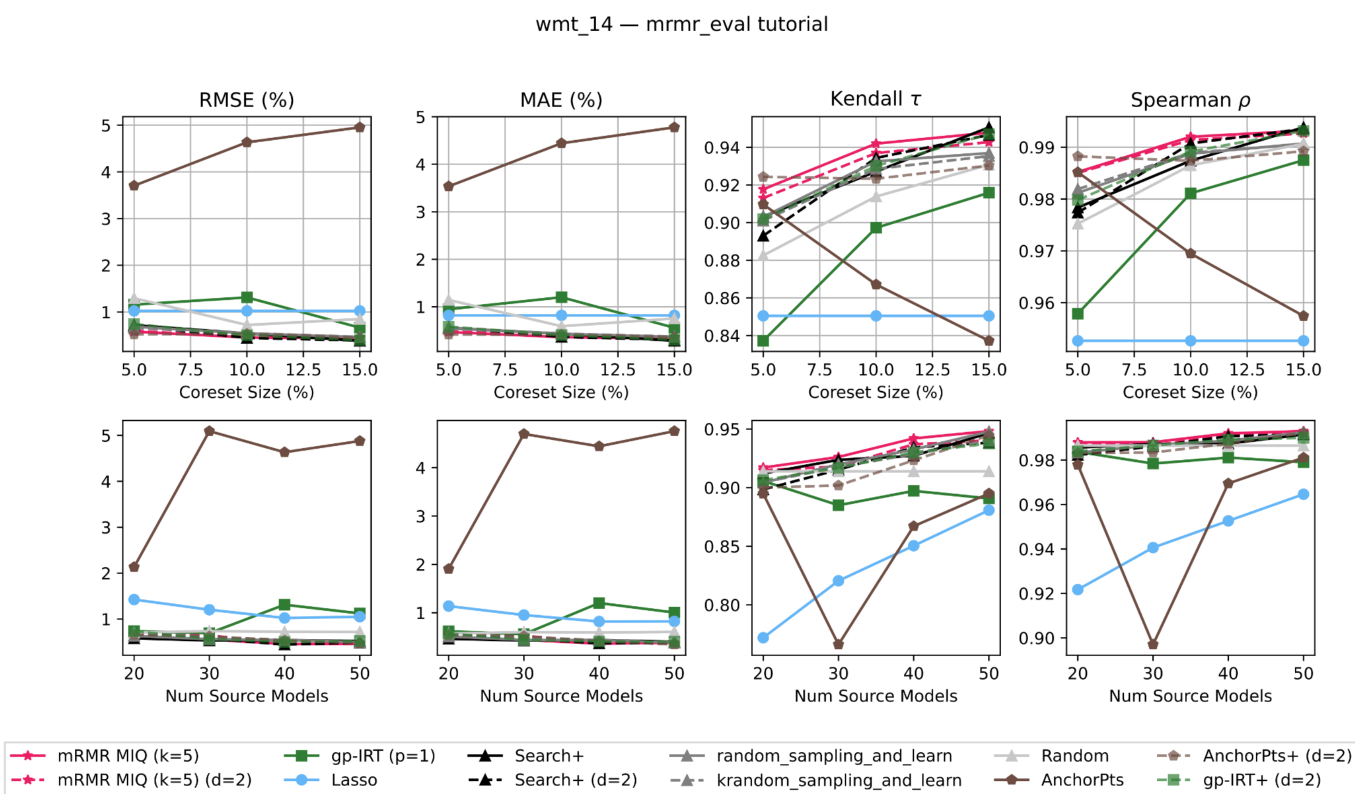
4. Thread C — efficient benchmarking

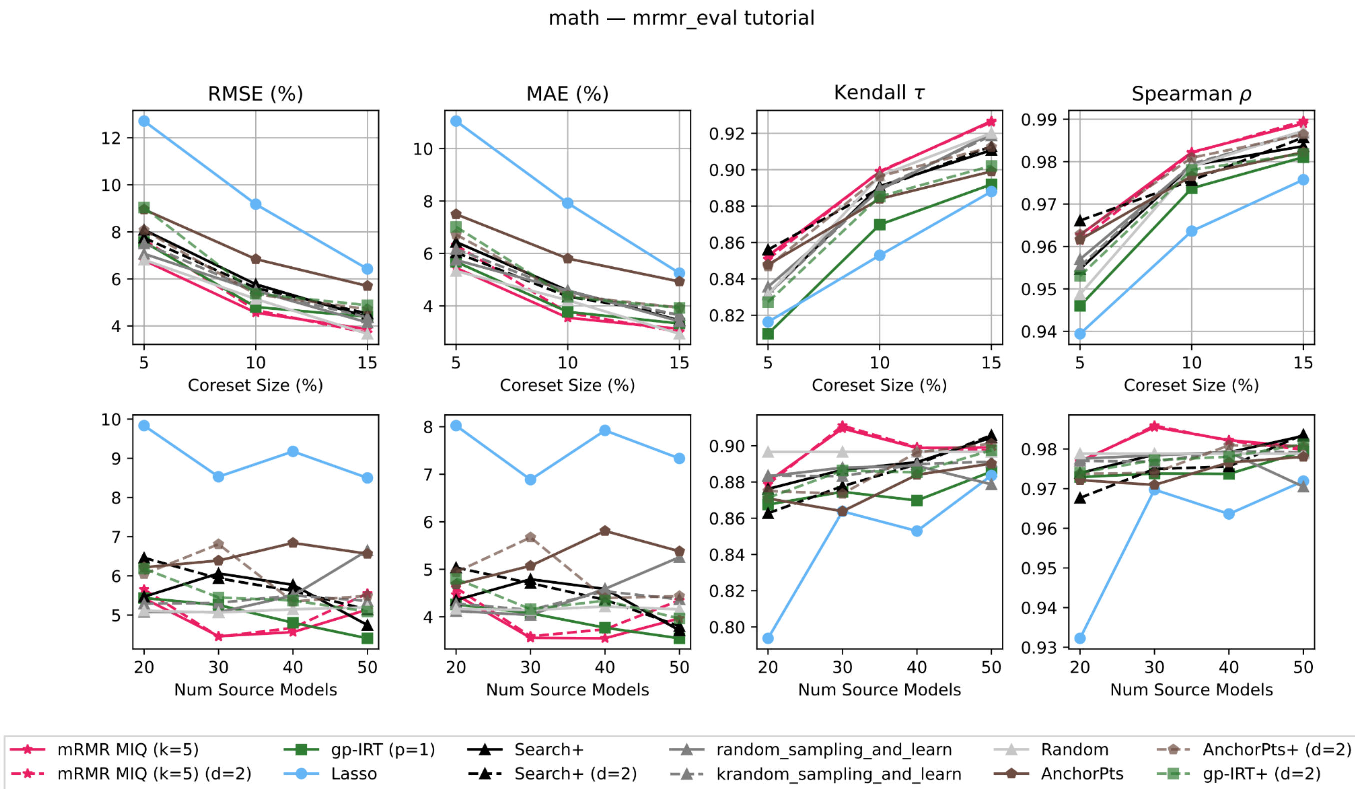
Ran 2026-06-02 -> 06-23, then paused when attention returned to reproduction. Results exist **only in the decks** and in `/data/dkps-embeddings` (19 GB of cached embeddings). Sibling repos `mrmr_eval` / `dkps` hold the code.

C1 — mRMR tutorial replication. Datasets `[legalbench, math, med_qa, wmt_14]`; reference models `[20, 30, 40, 50]`; coreset `[5, 10, 15] %`; 12 methods (mRMR MIQ, gp-IRT, Lasso, Search+, AnchorPoints±, random baselines, kernel variants). Metrics: RMSE, MAE, Kendall tau, Spearman rho.

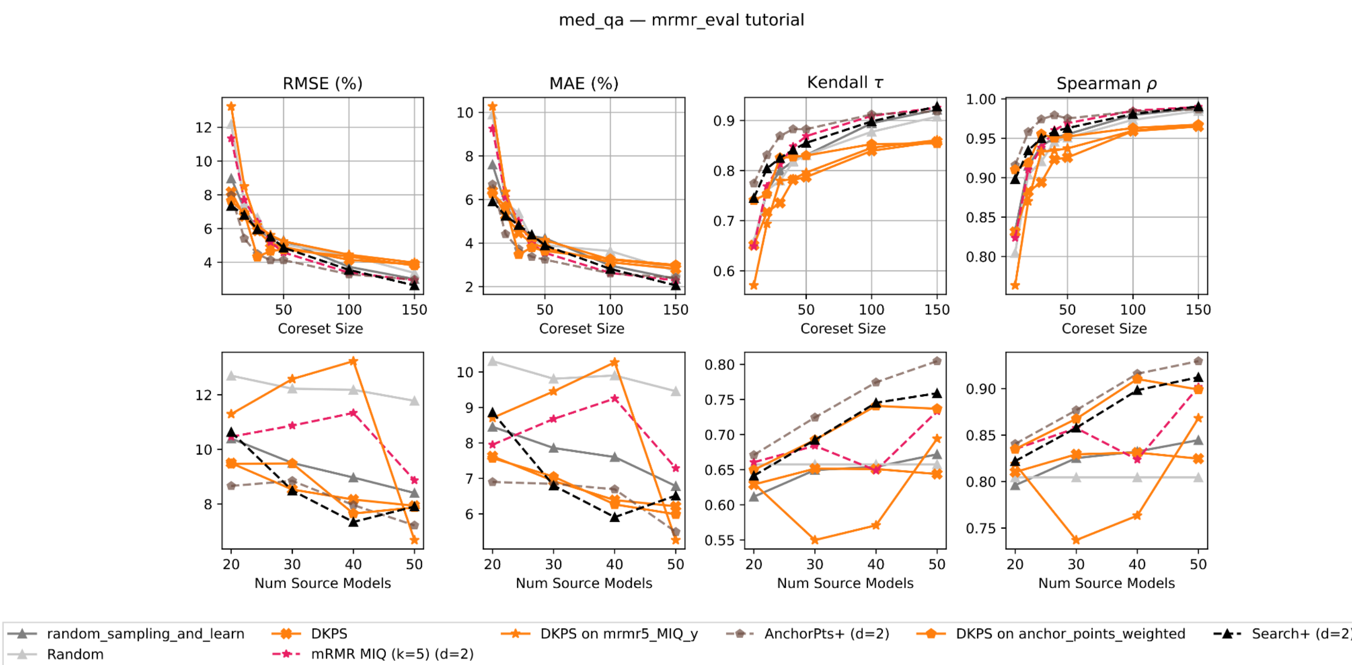



med_qa — mrmr_eval tutorial
RMSE (%)
MAE (%)
Kendall τ
Spearman ρ
Coreset Size
Num Source Models
mRMR MIQ (k=5)
mRMR MIQ (k=5) (d=2)
gp-IRT (p=1)
Lasso
Search+
Search+ (d=2)
random_sampling_and_learn
krandom_sampling_and_learn
Random
AnchorPts
AnchorPts+ (d=2)
gp-IRT+ (d=2)


wmt_14 — mrmr_eval tutorial
RMSE (%)
MAE (%)
Kendall τ
Spearman ρ
Coreset Size (%)
Num Source Models
mRMR MIQ (k=5)
mRMR MIQ (k=5) (d=2)
gp-IRT (p=1)
Lasso
Search+
Search+ (d=2)
random_sampling_and_learn
krandom_sampling_and_learn
Random
AnchorPts
AnchorPts+ (d=2)
gp-IRT+ (d=2)



C2 — DKPS vs baselines (06-23 deck, med_qa, DKPS one-hot + 8-D MDS):



Findings as recorded (deck 06-16 slide 12, 06-23 slide 6): - At 20 models / ~50 queries the feature-selection methods **beat** DKPS. - DKPS improves with more source models; the others do not necessarily. - mRMR is **extremely slow** on continuous-score benchmarks (wmt_14). - med_qa one-hot encoding is >5-D because not all models answer in-format — a representation problem, and one-hot exposes far less than text encoding. - Tutorial code runs each (method x coreset x n_models) cell only **3 times** — too few replicates to separate close methods.

Status: proposal, not result. The active-selection method (DKPS + mRMR), adaptive benchmarking, and the ensemble idea were designed and never run.

5. Everything that ran but produced no analysis

Ordered by cheapness of recovery. All raw artifacts are on disk.

#	What	Runs	What is missing	Cost
1	**Qwen3.5-9B-Base core grid**	72	virtual experiment + report	one command
2	**Judge analyses for 4 benchmarks x 4 judge sizes**	63 attempts	`30_analyze_judges.sh` per benchmark	minutes, zero GPU
3	`audit-falcon-7b-helm-grid`	41	any analysis at all	one command
4	`audit-gpt-oss-20b-core-grid`	40	any analysis at all	one command
5	**Repeatability pairs** (pythia/vicuna r1 vs r2)	12	reports never built	one command each
6	`audit-allenai-olmo-7b-mmlu-full` / `olmo-1-7-7b-full`	114 + 114	folded into combined? unclear	check
7	13B full-N HF-fp32 probe	0	oracle snapshotted, no cells run	~1 GPU-hour
8	Omni-MATH rejudge	0	never run; annotator never live-tested	GPU

6. Preservation status *(disk, verified today)*

store	raw run artifacts on this host	note
`audit-qwen-combined-full`	**703 / 703**	complete
`audit-allenai-olmo-combined-full`	**73 / 73**	complete
`audit-openai-gpt-oss-20b-from-spec-full`	**4 / 4**	complete (Jul 14)
`era-redpajama_3b-v0_{2_4,3_0}-*`	**16 / 16**	complete
`audit-*-ifeval-fp32` (7B, 13B)	**2 / 2**	complete (Jul 22)
`audit-qwen35-9b-base-vllm-full`	**72 / 72**	complete
deployment-match sweeps	19 scored + 46 sweep JSON	complete
open-judge snapshots + attempts	6 snapshots, 75 attempts	complete
Total	**1,634 runs**	

Nothing is hashed or bundled. Every experiment directory pairs 1:1 `run_spec.json` <-> `scenario_state.json` — I checked all 74. See §8.1 for why this contradicts the master reference.

7. Figure index

All figures in `figures/`, copied from the report trees. Regenerate any of them with the `redraw_plots.sh` next to its source.

file	source
`{olmo,qwen,gpt-oss-20b}_score_drift_headline.png`	`.../level_001/aggregate_score_diff/`
`{olmo,qwen,gpt-oss-20b}_agreement_curve.jpg`	`.../aggregate-summary/`
`{olmo,qwen,gpt-oss-20b}_agreement_curve_per_metric.jpg`	"
`{olmo,qwen,gpt-oss-20b}_reproducibility_buckets.jpg`	"
`{olmo,qwen,gpt-oss-20b}_sankey_reproducibility.jpg`	"
`{olmo,qwen}_sankey_scope_to_analyzed.jpg`	"
`{olmo,qwen,gpt-oss-20b}_failure_taxonomy.jpg`	"
`olmo_drift_ifeval.png`	`.../aggregate_score_diff_per_metric/`
`era_redpajama_v0{24,30}_agreement_curve.jpg`	era virtual experiments
`e2e-phi2-{hf,vllm,incomparable}_core_metric_report.png`	e2e core reports
`threadB_mrmr_*.png`, `threadB_dkps_*.png`	decks 06-16 / 06-23 (exist nowhere else)

Not copied but available: ~8,900 further plots, notably `qwen-models-combined` (6,129) and `olmo-models-combined` (1,555), including per-metric drift heatmaps and per-run core-metric reports; and the interactive `.html` sankeys beside every `.jpg`.

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8. Discrepancies found while compiling this

Read this section before citing anything above. Four disagreements between the master reference and what is on disk today. Three of them make the project's position *better* than the written record says.

8.1 The flagship stores are NOT pruned on this host — preservation is cheap

The 2026-07-15d audit reported `gpt-oss-20b-from-spec` and `olmo-models-combined` as **"ABSENT (pruned; 0 scenario_state)"** and concluded that "regenerate" means **re-run**. That conclusion is carried into the master reference, the claim ledger, and the strategy addendum as the most urgent open item.

On this host today, both stores are complete: GPT-OSS 4/4 scenario_state, OLMo-combined 73/73, and 1,634 runs in total across all 74 experiment directories, every one pairing 1:1 with its `run_spec.json`. The GPT-OSS artifacts are dated 2026-07-14 09:29–13:19 — i.e. they were on disk *before* the audit ran.

The likely explanation is that the audit inspected a different root (it ran on `aiq-gpu`; this is the workstation mirror, which received a 315 GB pull on 2026-07-22). Either way the **actionable conclusion inverts**: preservation is copy-and-hash from this host, not a re-run campaign. This should be checked on `aiq-gpu` and the store ledger corrected.

8.2 The "59 % classic-track" figure is not supported by the inventory

The master reference carries ~59 % of the corpus as classic-track, flagged `internal-estimate`. The inventory says **classic is 8,377 / 34,512 = 24.3 %** of run records (159 of 1,109 selected = 14.3 %). Either the 59 % refers to a different denominator or it is wrong. Do not publish it before regenerating.

8.3 The marin cross-family confirmation already exists

The master reference records the marin-8b registered prediction as *untested* (the first sweep was void — the protocol resolver silently defaulted to completions). The chat-protocol rerun **exists on disk and passed**: fp32 MATCH 1.000, fp16 0.909, auto/bf16 PARTIAL 0.517. That is the "unpinned dtype ⇒ fp32 wins" prediction confirmed on a **second family**, which is exactly what Task A of the task queue was created to obtain. Also note the 7B **full-N** probe ran (§8.4) — the master reference lists it as an optional open item.

8.4 "541/541 byte-exact" should be 539/541

The full-N probe's `ranking.txt` reports `exact 1.00`, and the journals predict 541/541. Direct byte comparison of completions against the oracle gives **539/541 (99.63 %)** for every complete cell. Two long-form instances diverge mid-generation. At n=32 it genuinely is 32/32 — the two misses are outside that sample. The claim survives comfortably (chance agreement is ~0) but the number in the paper must be 539/541, and the two divergences are worth a sentence: they are exactly the near-tie token flips the thesis predicts, showing up *within* the recovered configuration.

Also worth noting, in the project's favour: the OLMo headline table on disk shows MMLU olmo-7b **0.295/0.287**, the corrected dedupe value — not the stale 0.295/0.144 the master reference says is still on disk. That store was regenerated 2026-07-14 12:27 local.

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9. Suggested next steps

Free (no GPU), highest value first 1. Build the Qwen3.5 report (§A9) — a complete extension result is sitting in raw form. 2. Re-run the judge analyses for the 4 newer benchmarks and 4 smaller judges (§B3) — the agreement/kappa numbers for two-thirds of the grid do not exist yet. 3. Copy + hash everything (§6, §8.1) — and correct the store ledger. 4. Build reports for falcon-7b, gpt-oss-core-grid, and the repeatability pairs (§5). 5. Regenerate the corpus denominator from a checked-in manifest (§8.2).

Cheap GPU 6. Finish the 13B full-N probe (§A7) — one cell, ~1 GPU-hour. 7. Cross-family sweep: pythia-6.9B, vicuna-7B, granite-4.0-micro as treatment, gemma-2-9b-it (pinned bf16) as bidirectional control. **marin is already done and confirms** (§8.3).

The real gap 8. The prospective census — frozen sampling rule, frozen diagnostic ladder, outcome buckets defined in advance. Everything above is one deep case (A8) plus broad agreement statistics (A3/A4/A5). Neither is a population claim, and no amount of further per-experiment work turns them into one. 9. Candidate expansion for the judge study — every endpoint needs a *set* of candidates, and the corpus already holds official responses and judgments for the whole leaderboard, so this costs judge inference only.